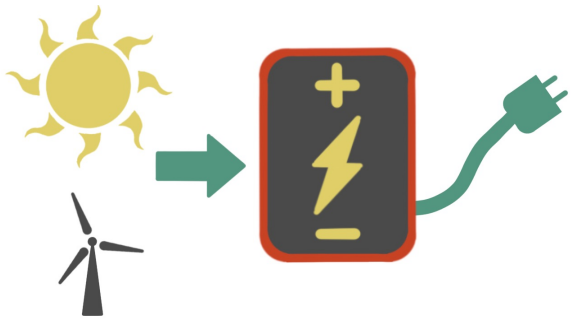
Value of Accuracy



Ania's BMS Course

by Ania Mitros, PhD

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Value of Accuracy : Video 1

Video 1
- Systems engineering: Quantifying design variables in $

Video 2
- Cell voltage measurement accuracy ($/mV)
  - Review
  - Calculate value in $
    - 1st order
    - Better analysis

Video 3
  - Accuracy on BMS chip datasheets

Video 4
- Battery current measurement accuracy ($/A)

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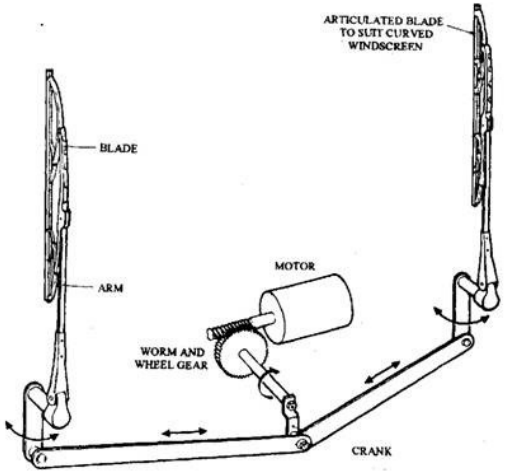
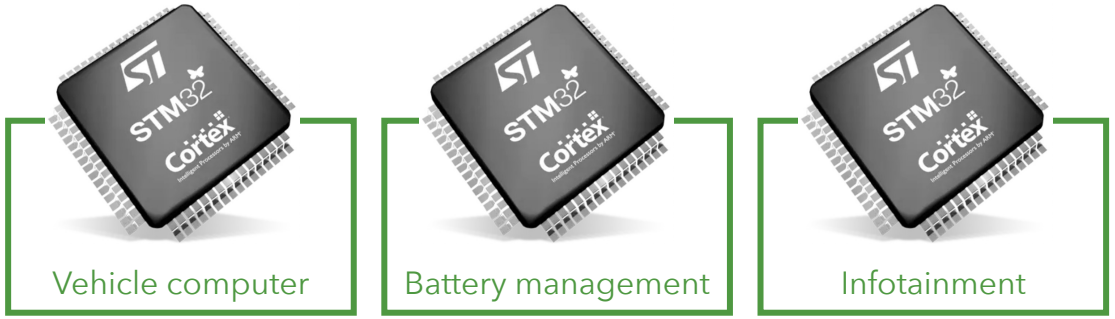
System engineering: Trade-offs





Weight
(mechanical trade–off)

Energy usage
(electrical trade–off)

Car drawing by Jozef Mikulcik from Pixabay

Windshield wiper from: http://what-when-how.com/automobile/windscreen-wipers-and-washers-automobile/

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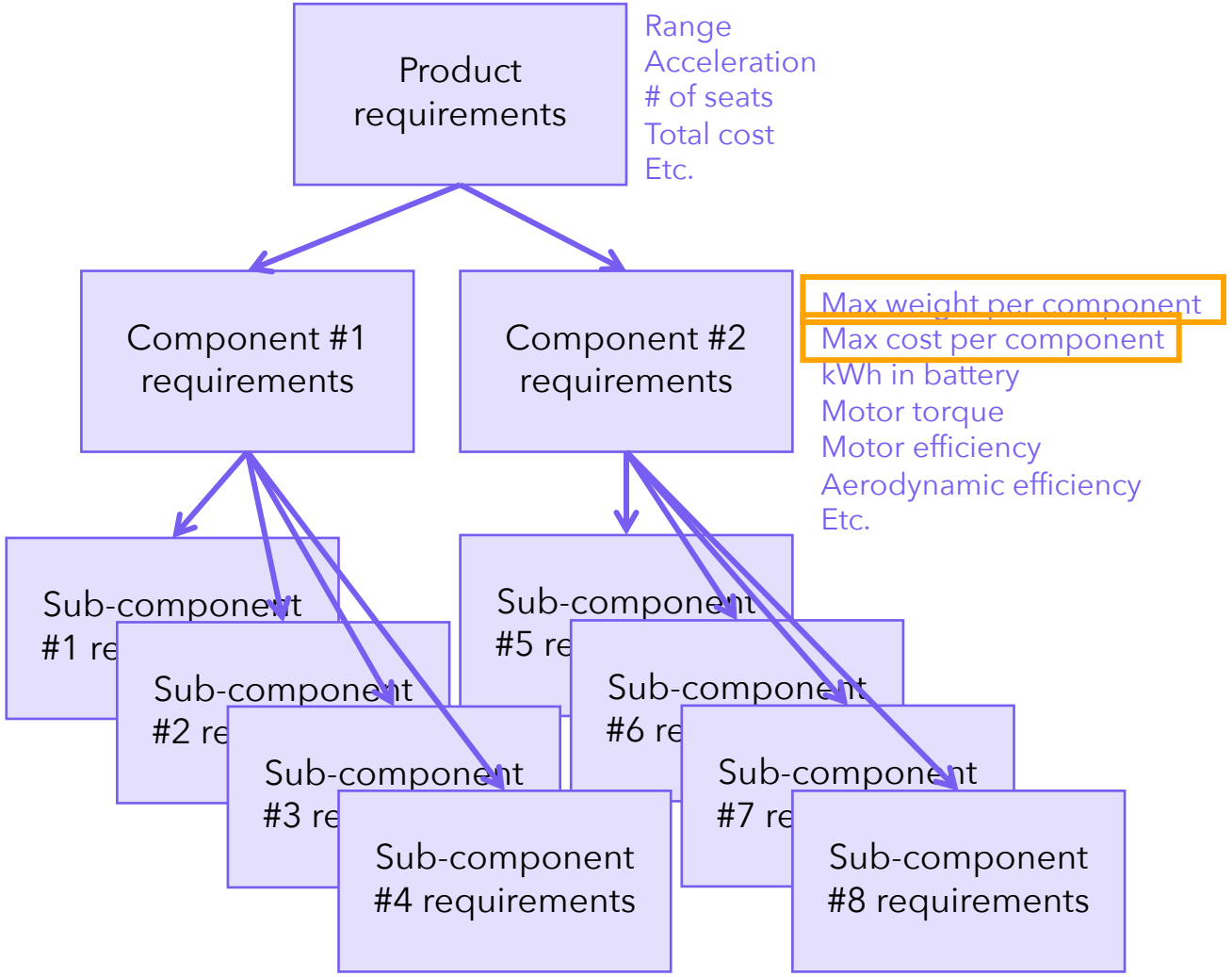
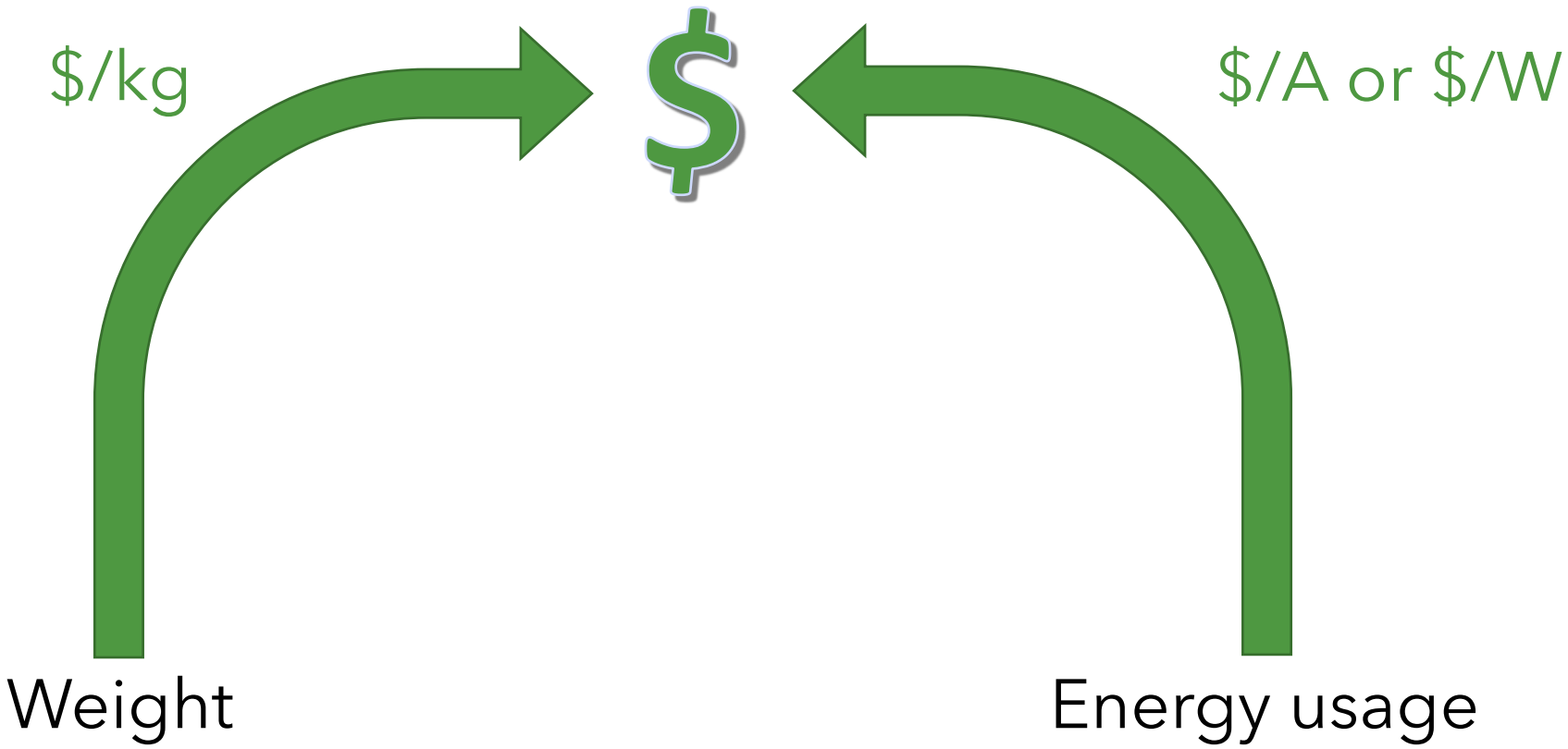
Traditional approach



Common target



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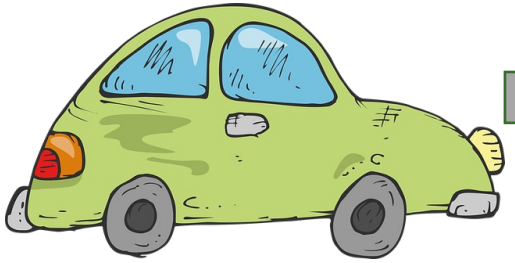
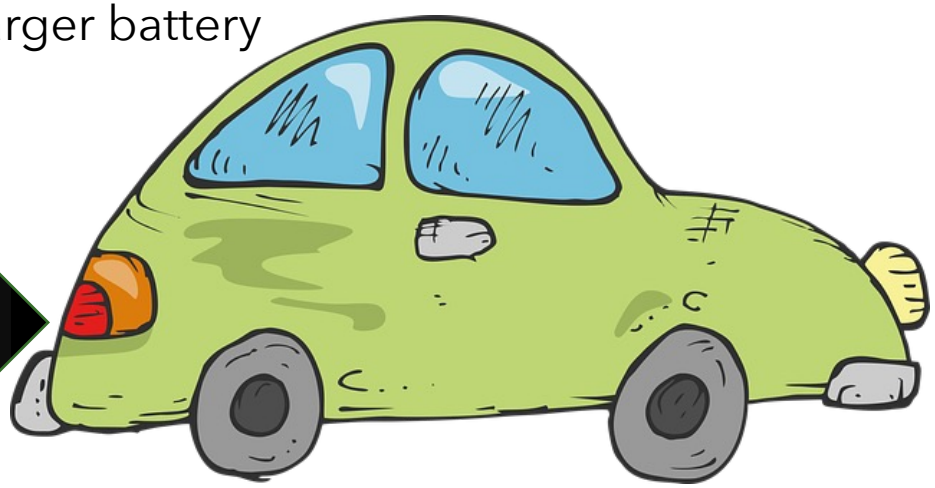
Calculate $/kg for an electric vehicle

- Same range → larger battery
- Stronger chassis
- Stronger brakes
- Etc.



Lighter → Heavier



Quantify: How much does weight affect total vehicle cost?

Consider battery cost

Assume: 10% weight reduction → 9% range improvement

Assume: $15,000 battery cost.

Assume: 1800 kg vehicle weight.

$$\frac{\$}{kg} = \frac{9\%}{10\%} \times \frac{(\$15{,}000)}{1800\ kg} = \$7.50/kg$$

More on weight vs range: https://www.teslasiliconvalley.com/blog//ev-weight-impact-on-range

Car drawing by Jozef Mikulcik from Pixabay

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Apply $/kg to decisions

$7.50 per kg

40g for $1 vs 38g for $1.03

2g $0.03

$$\frac{\$}{kg} = \frac{(\$1.03 - \$1.00)}{0.040kg - 0.038kg} = \$15.00/kg$$

$15/kg > $7.50/kg
Not worth it.
Buy the less expensive bolt!

$20 stamped part, weighing 100g.
A process change could reduce weight by 4%, with 0.1% worse yield.

$$\frac{\$}{kg} = \frac{(\$20 \times .001)}{0.100kg - 0.096kg} = \$5.00/kg$$

$5/kg < $7.50/kg
Make the process change!
Increase subcomponent cost for system-level savings

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Value of Accuracy : Video 1

Video 1
- Syst s in $

Video 2
- Cell
  - R
  - C
  - A

Video 3

Video 4
- Batt

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Value of Accuracy : Clip 2

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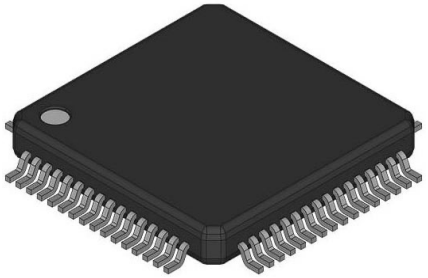
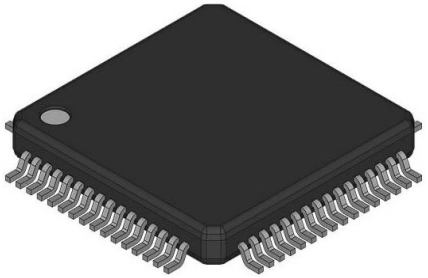
Value of cell voltage measurement accuracy



vs



±7mV

±2mV

Should you pay more for the more accurate chip?
How much more?

$7

$7

$10 $20

$??

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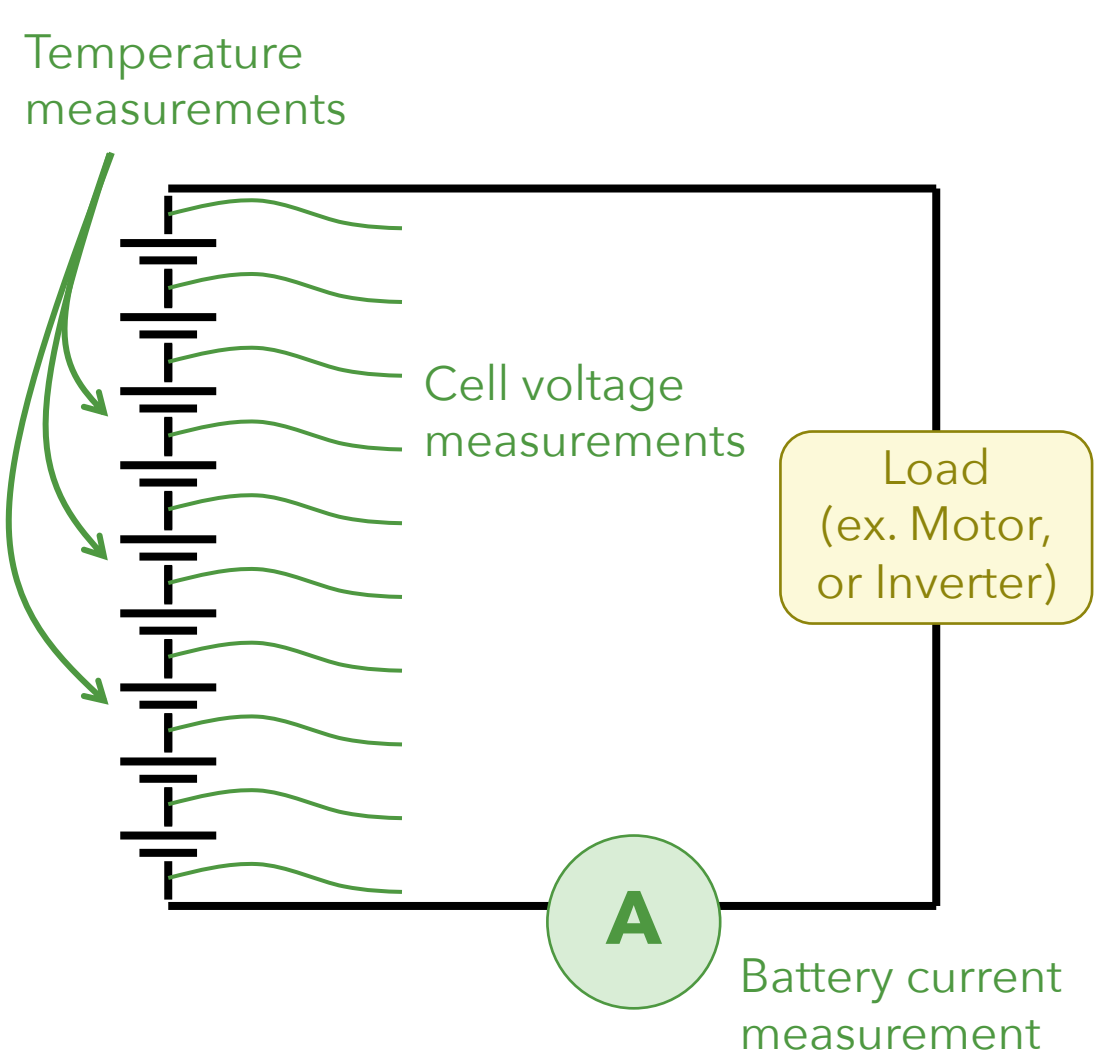
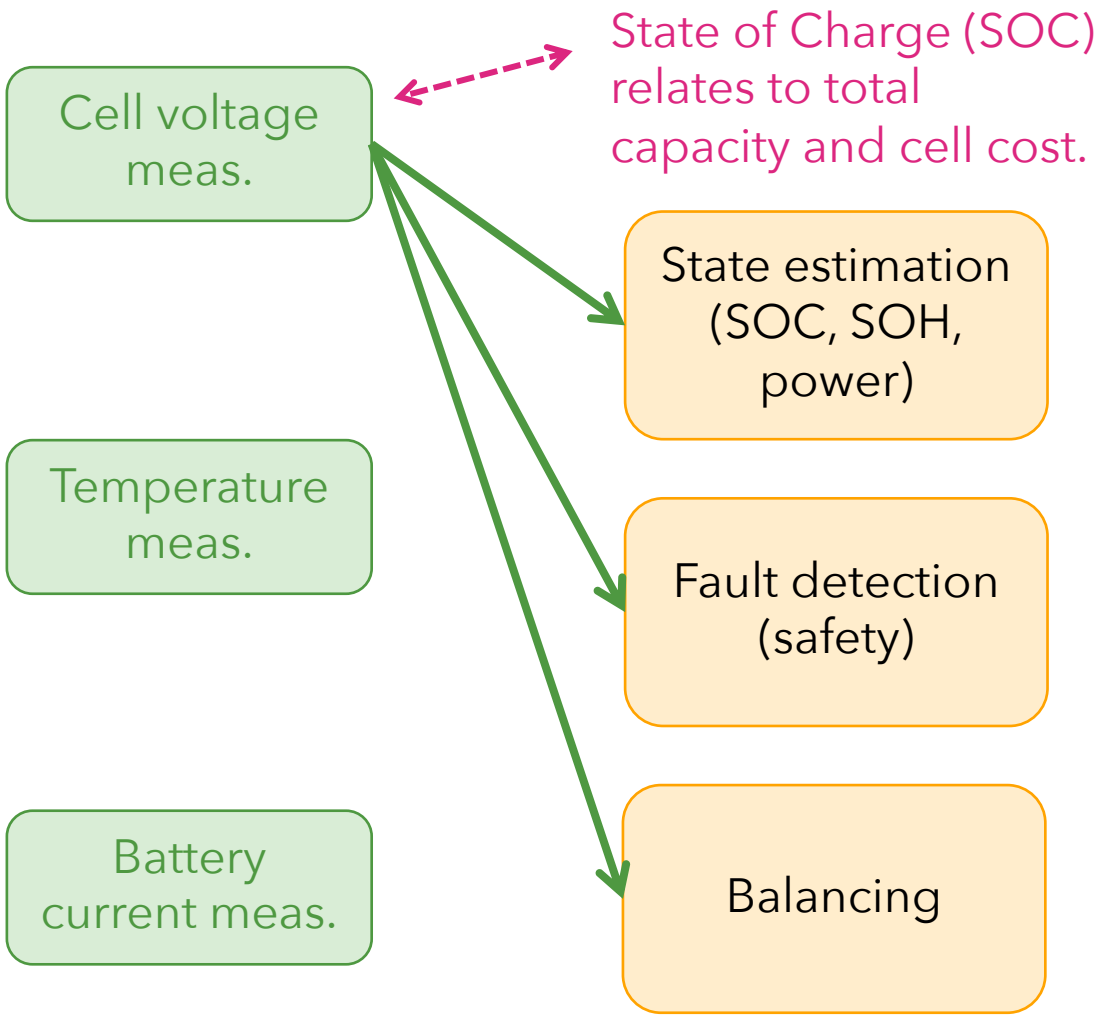
How we use cell voltage measurement





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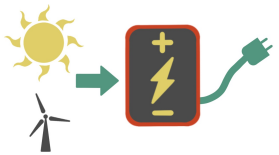
Value of Accuracy



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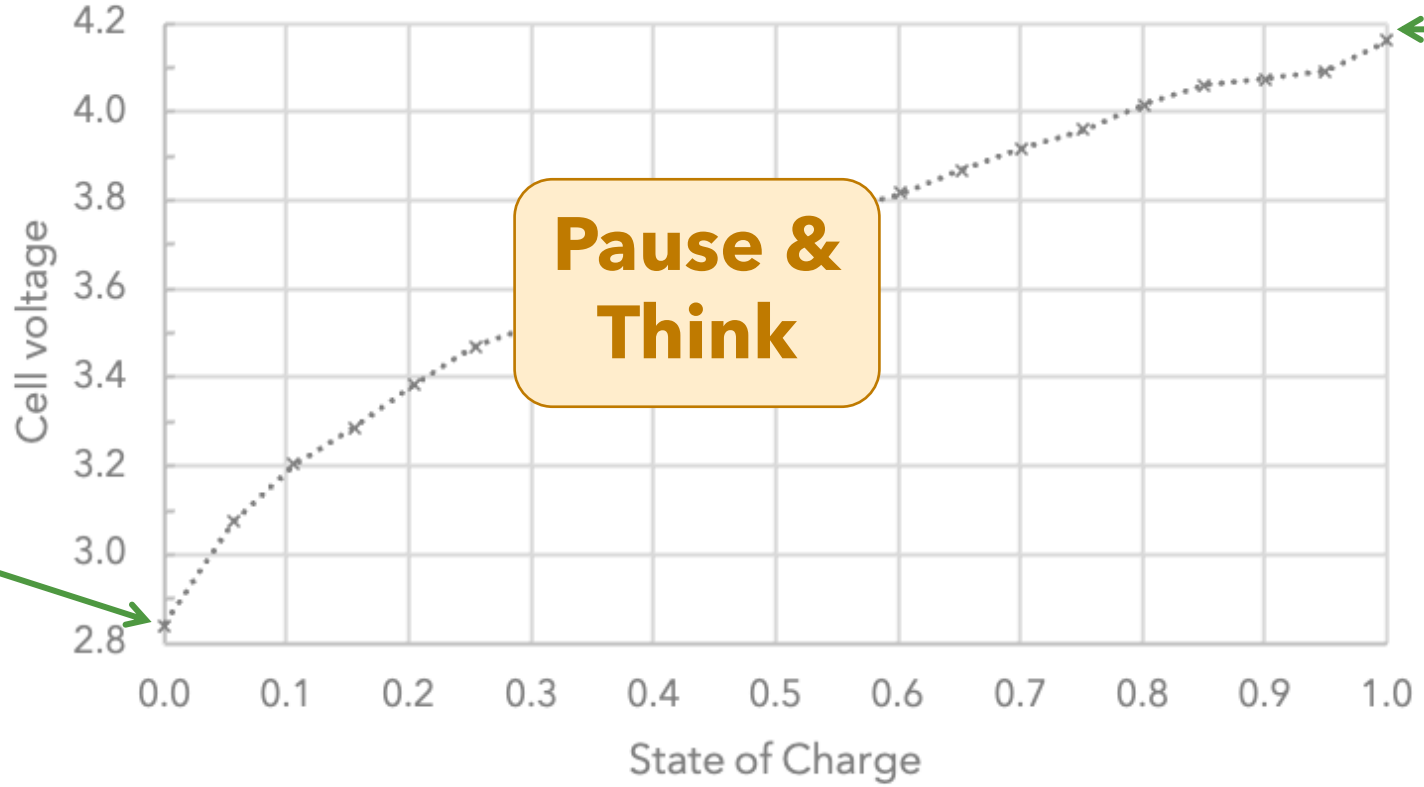
How does voltage accuracy matter?

Ground truth: Can never estimate SOC better than from cell voltage.

If we measure **too high**, we stop **charge** prematurely.



If we measure **too low**, we stop **discharge** prematurely.

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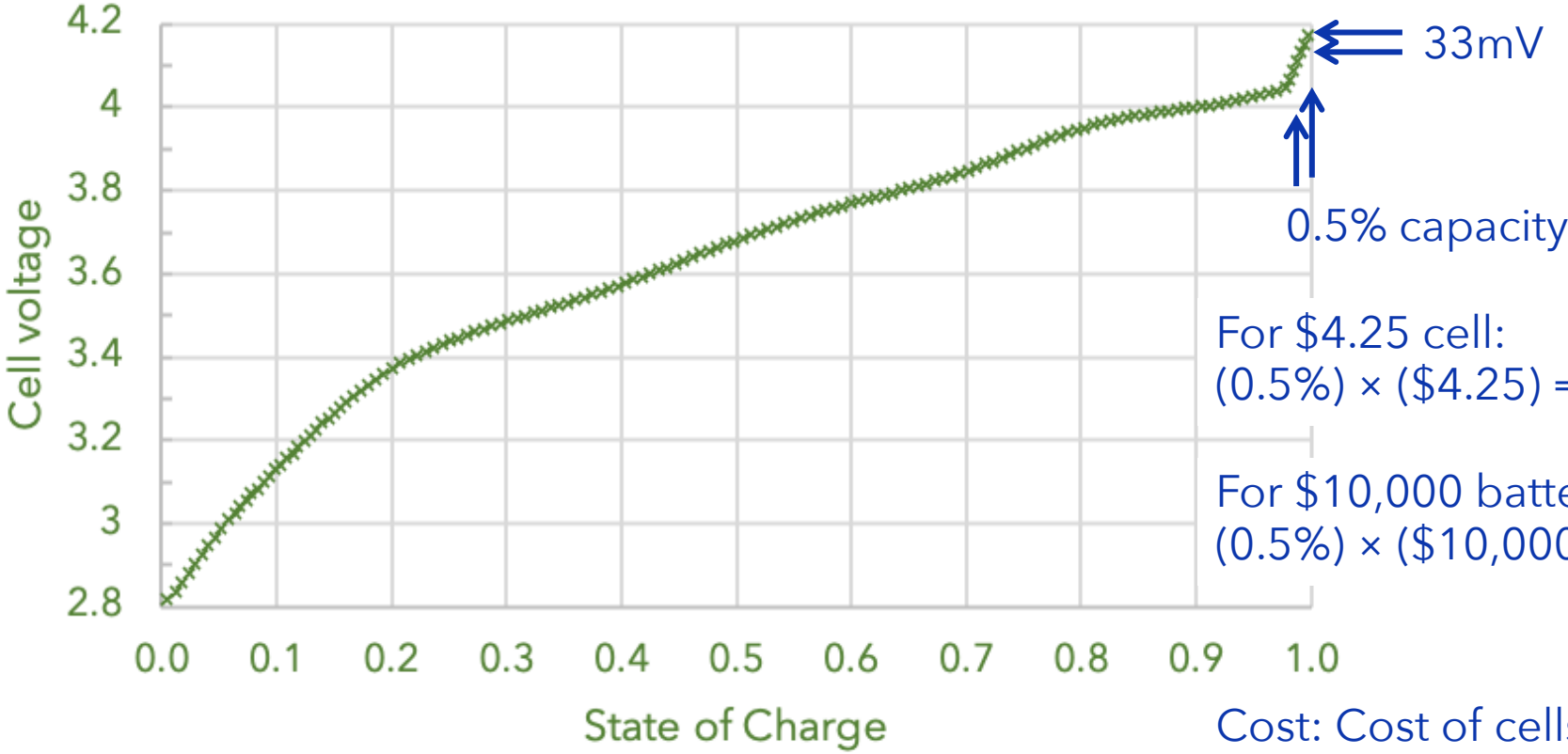
Quantify value of accuracy in $ per mV

Pause & Think



For $4.25 cell:
(0.5%) × ($4.25) = $0.02

For $10,000 battery:
(0.5%) × ($10,000) = $50

Cost: Cost of cells added to compensate for error

$$\text{Cost in \$} = \frac{\cancel{[error\ in\ mV]}}{\left[slope\ in\ \frac{mV}{\%SOC}\right]} \times [battery\ cost\ in\ \$]$$

Error in mV

$$\text{\$ per mV} = \frac{[battery\ cost\ in\ \$]}{\left[slope\ in\ \frac{mV}{\%SOC}\right]}$$

System metric

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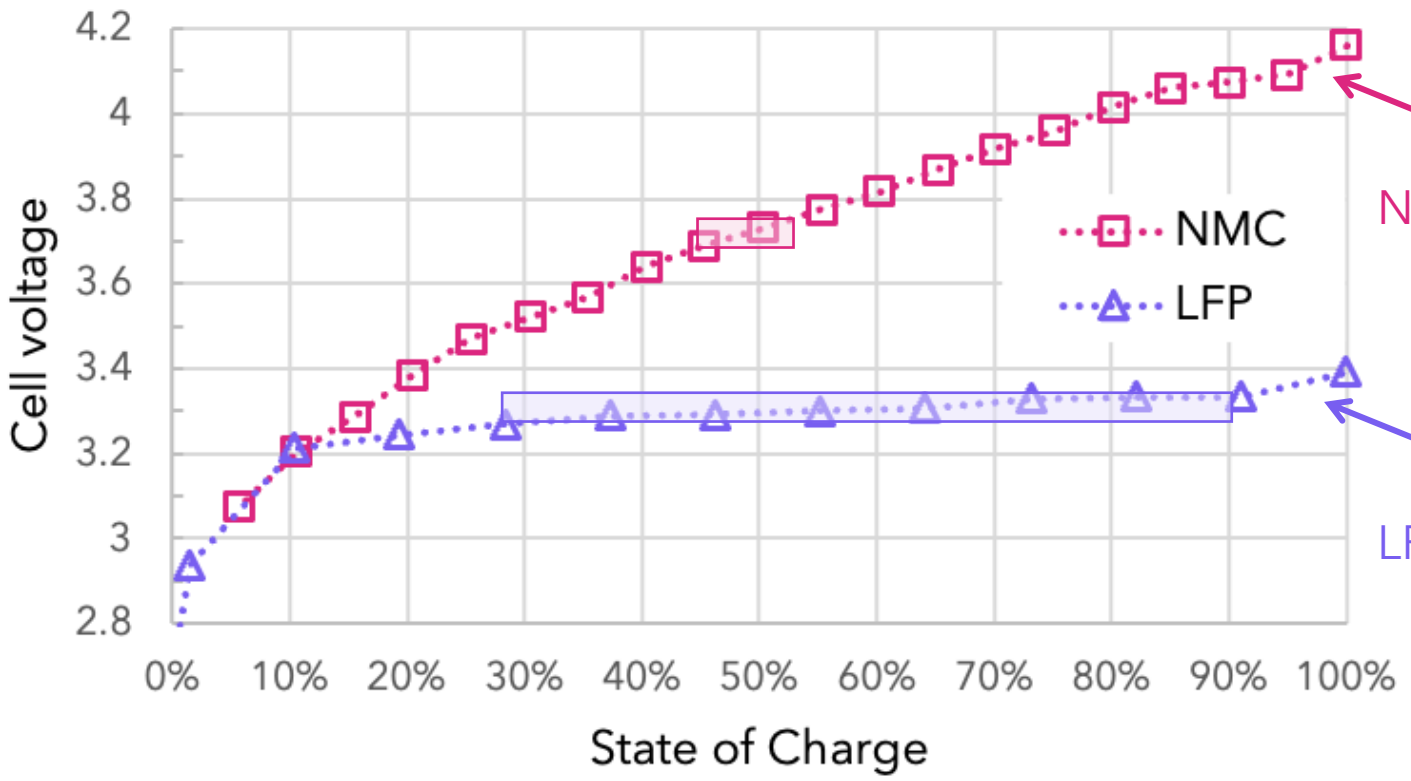
Value of accuracy varies across batteries

For which cell is accuracy more valuable?



1.42V / 100% SOC

NMC $/mV = $\frac{[battery\ cost\ in\ \$]}{[X]}$

0.67V / 100% SOC

LFP $/mV = $\frac{[battery\ cost\ in\ \$]}{\left[\frac{1}{2}X\right]} = \mathbf{2}\times[NMC\ \$/mV]$

$ per mV = $\frac{[battery\ cost\ in\ \$]}{\left[slope\ in\ \frac{mV}{\%SOC}\right]}$

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First-order analysis

Value of cell voltage measurement accuracy depends on the slope of the OCV vs SOC curve at:

- Top of charge, where it may limit charging, or
- Bottom of charge, where it may limit discharging.

$$\$ \text{ per mV} = \frac{[battery\ cost\ in\ \$]}{\left[slope\ in\ \frac{mV}{\%SOC}\right]}$$

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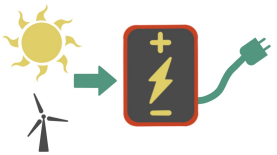
Value of Accuracy



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  - Accuracy on BMS chip datasheets

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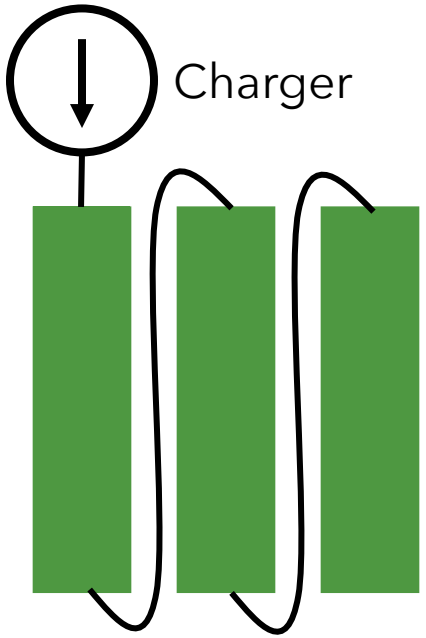
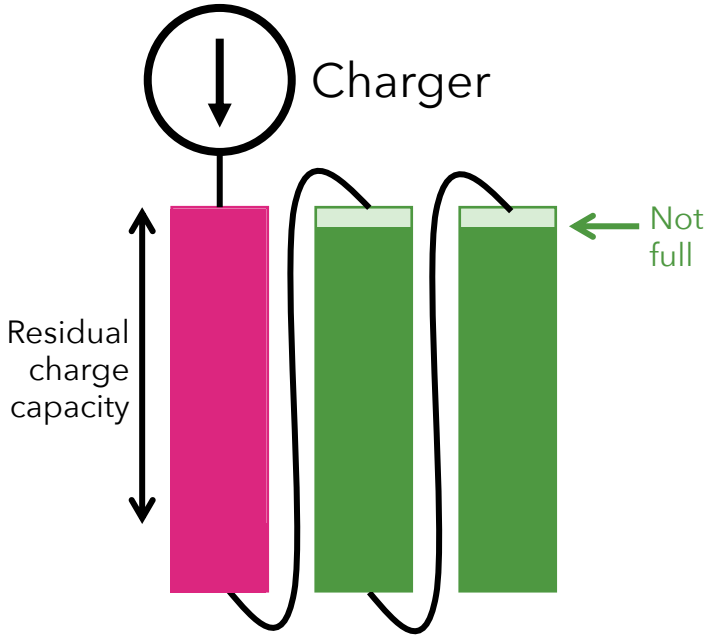
Better analysis involves balancing

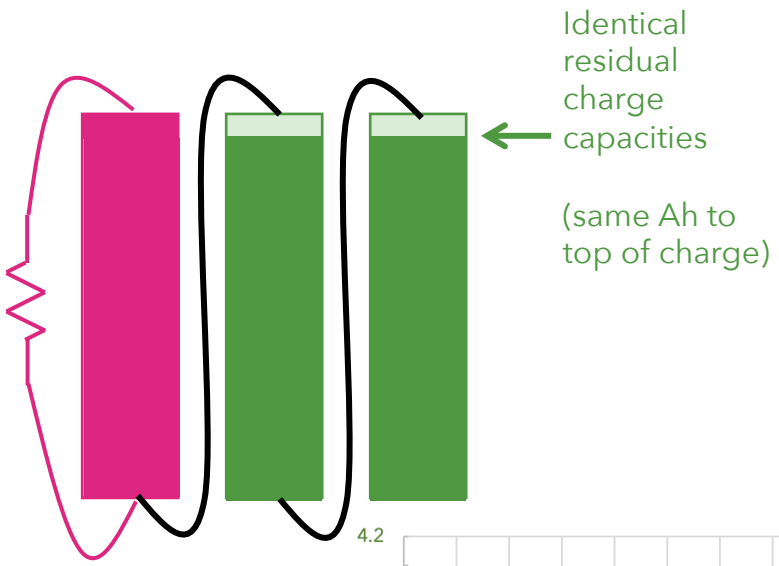
Quick review of balancing



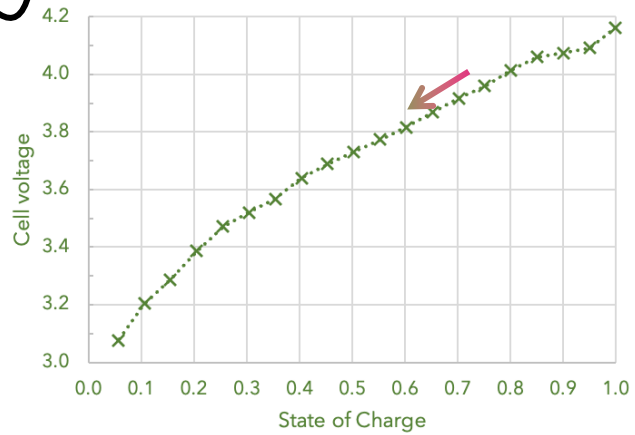
Ideally: All cells reach top of charge together



Reality without balancing



Balancing



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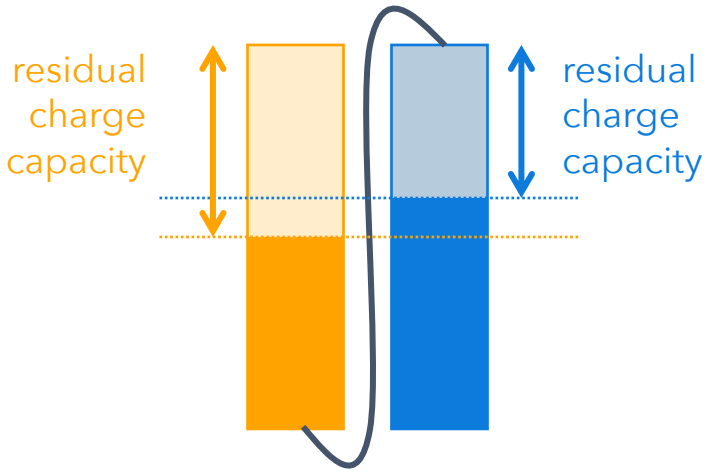
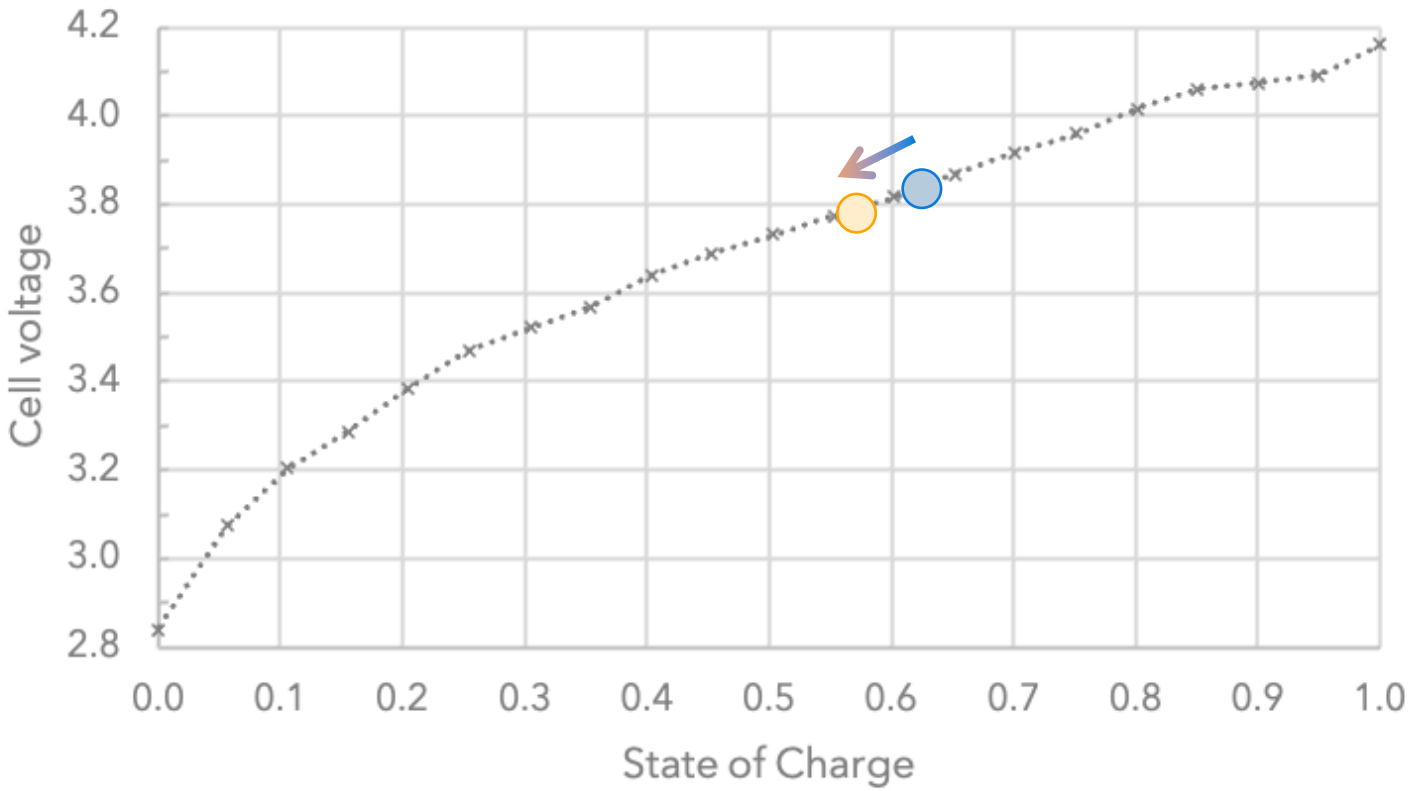
Better analysis centers on balancing





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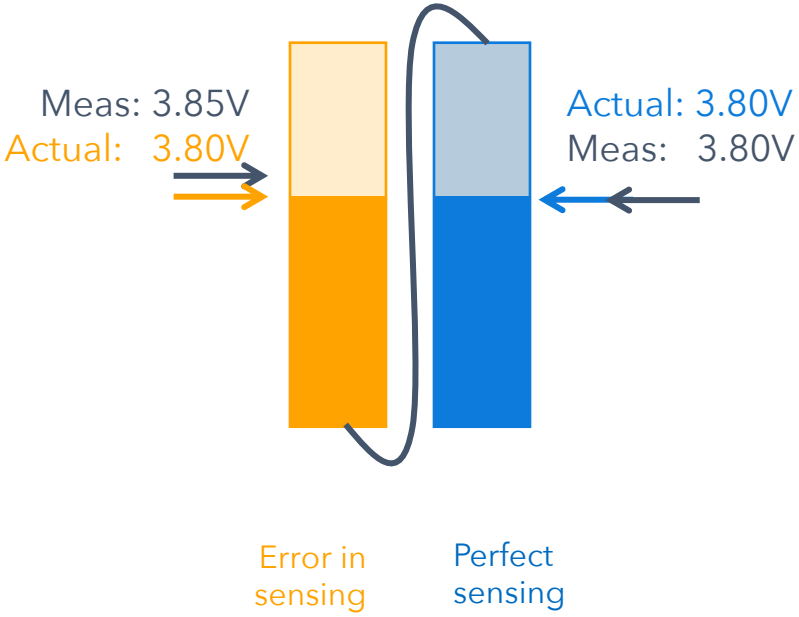
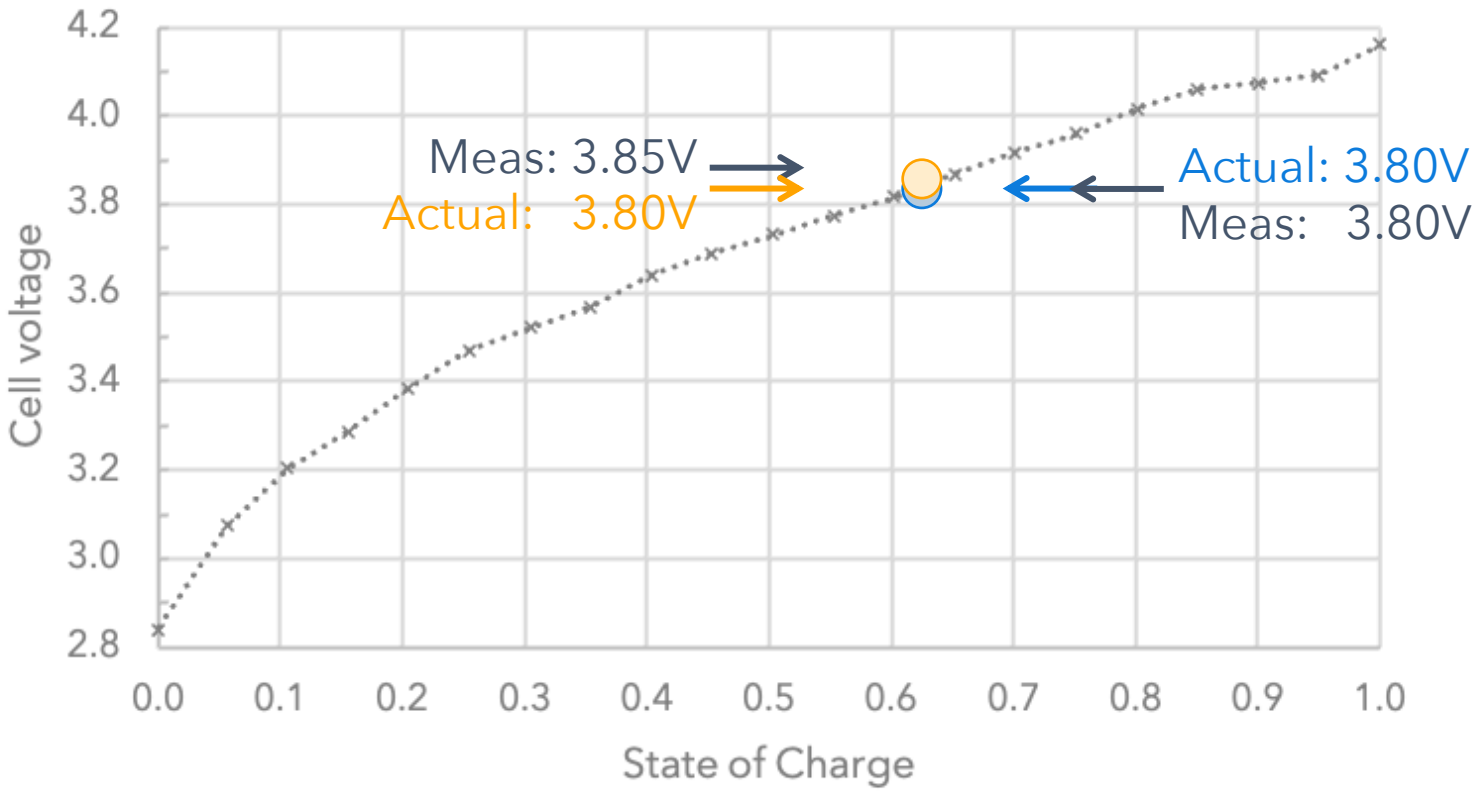
Impact of accuracy on balancing





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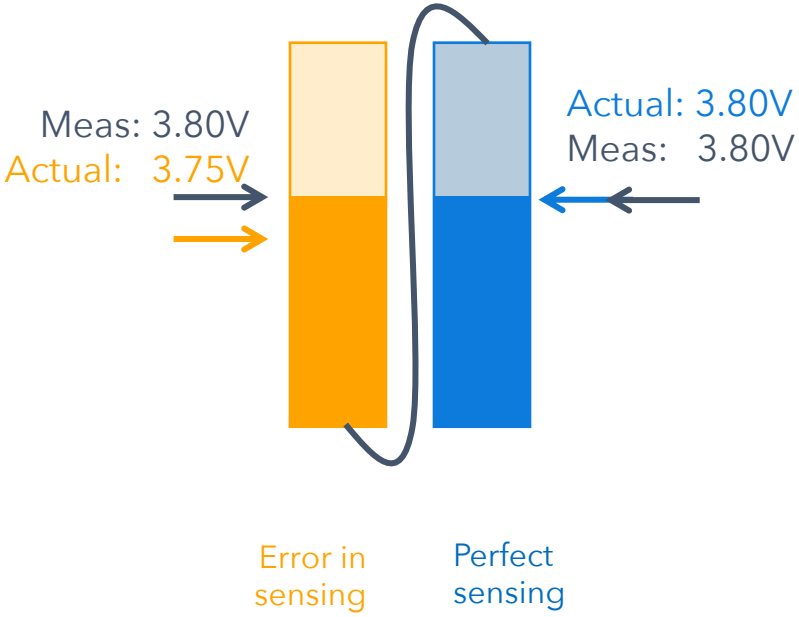

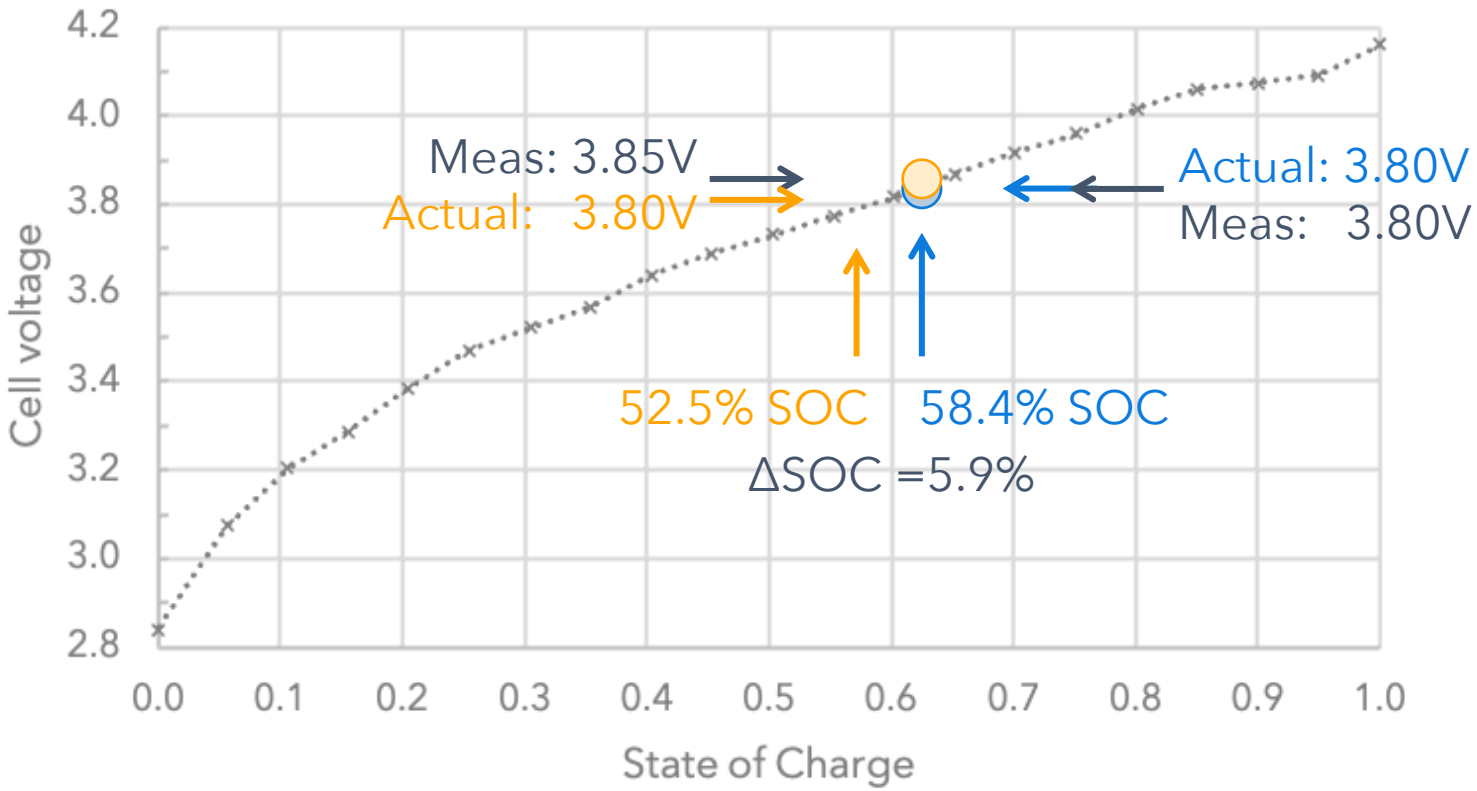

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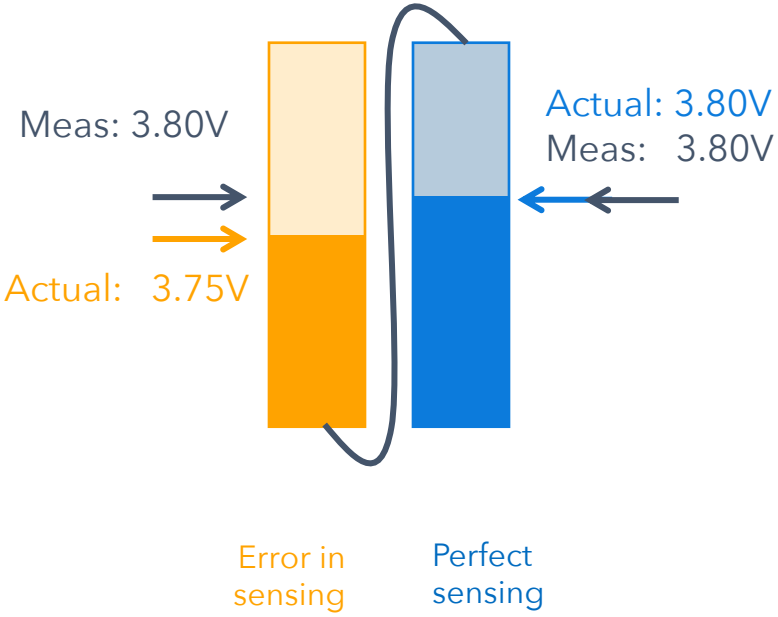

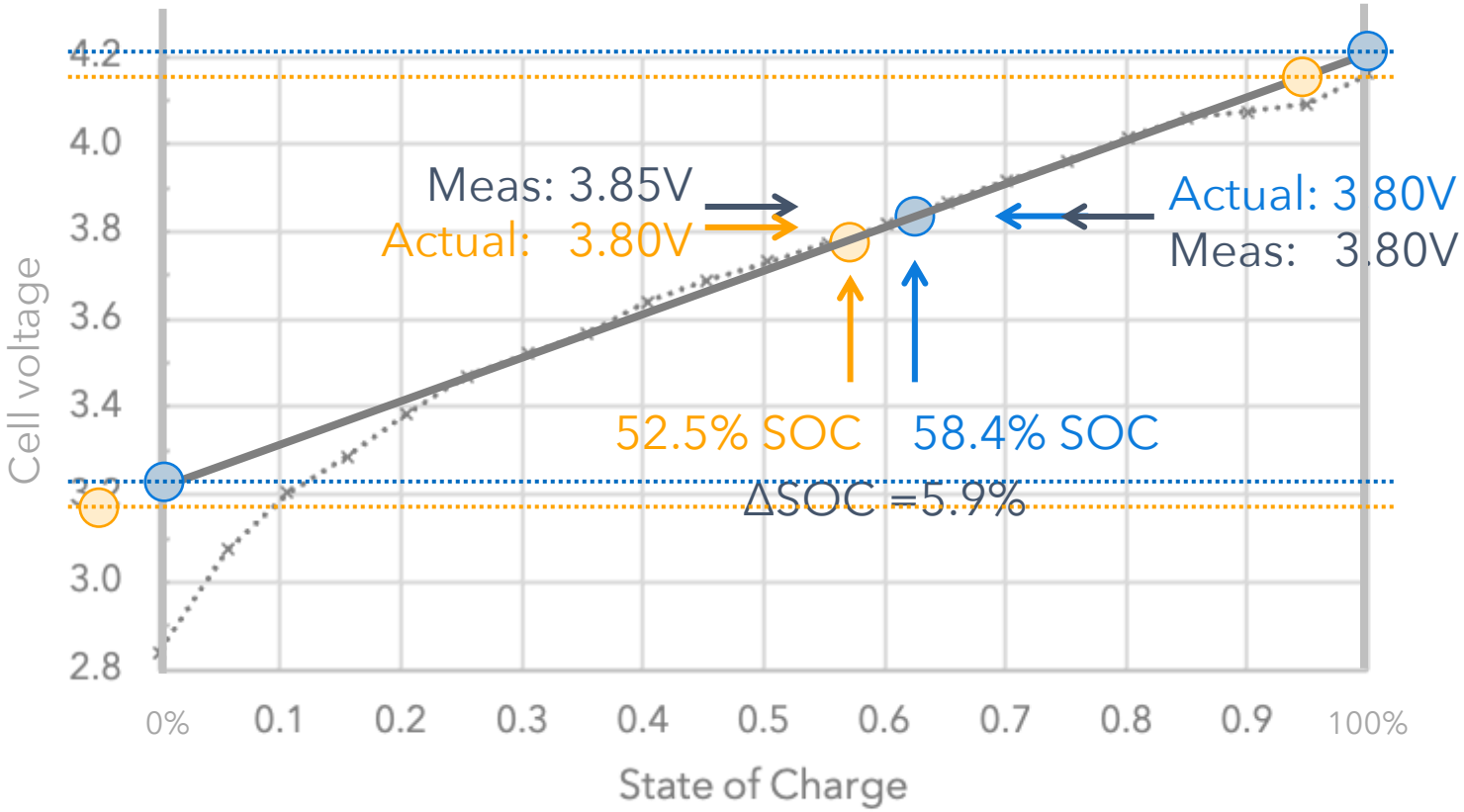

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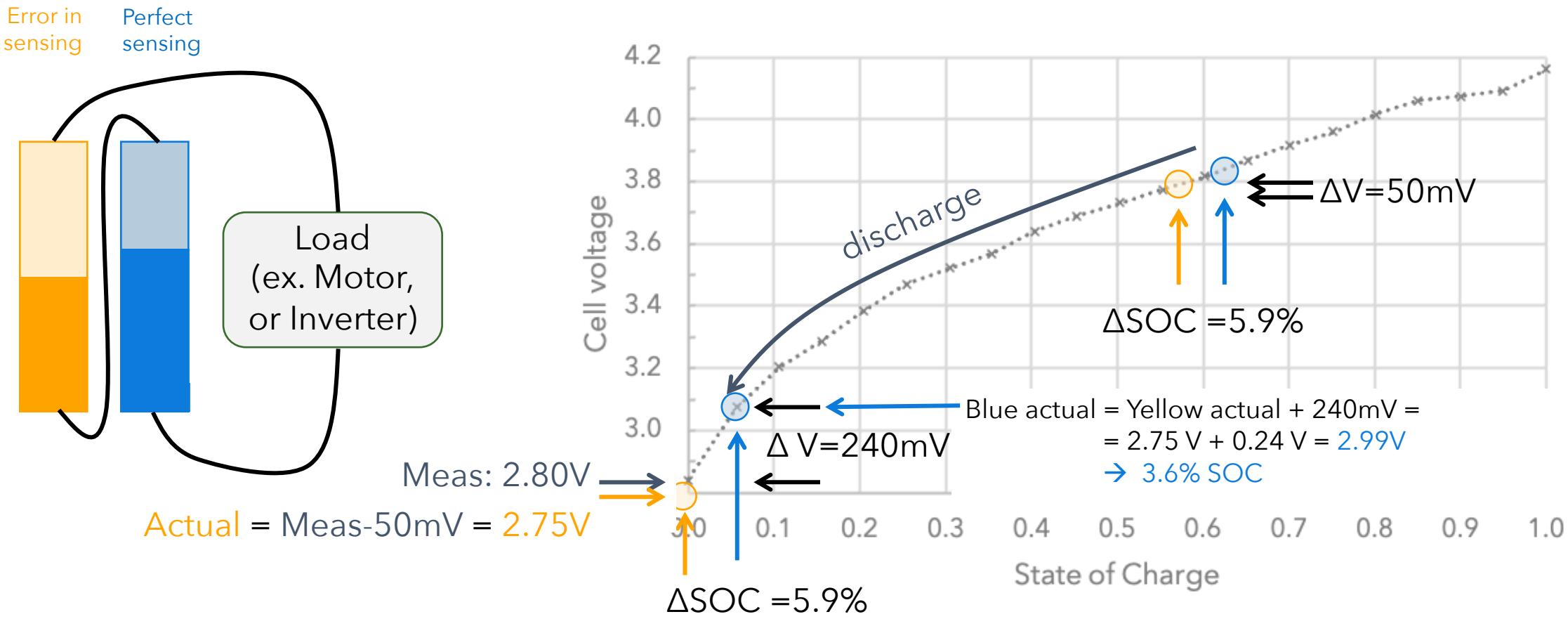
Imbalance impacts actual residual capacity

We lose 3.6% of battery capacity due to error in balancing



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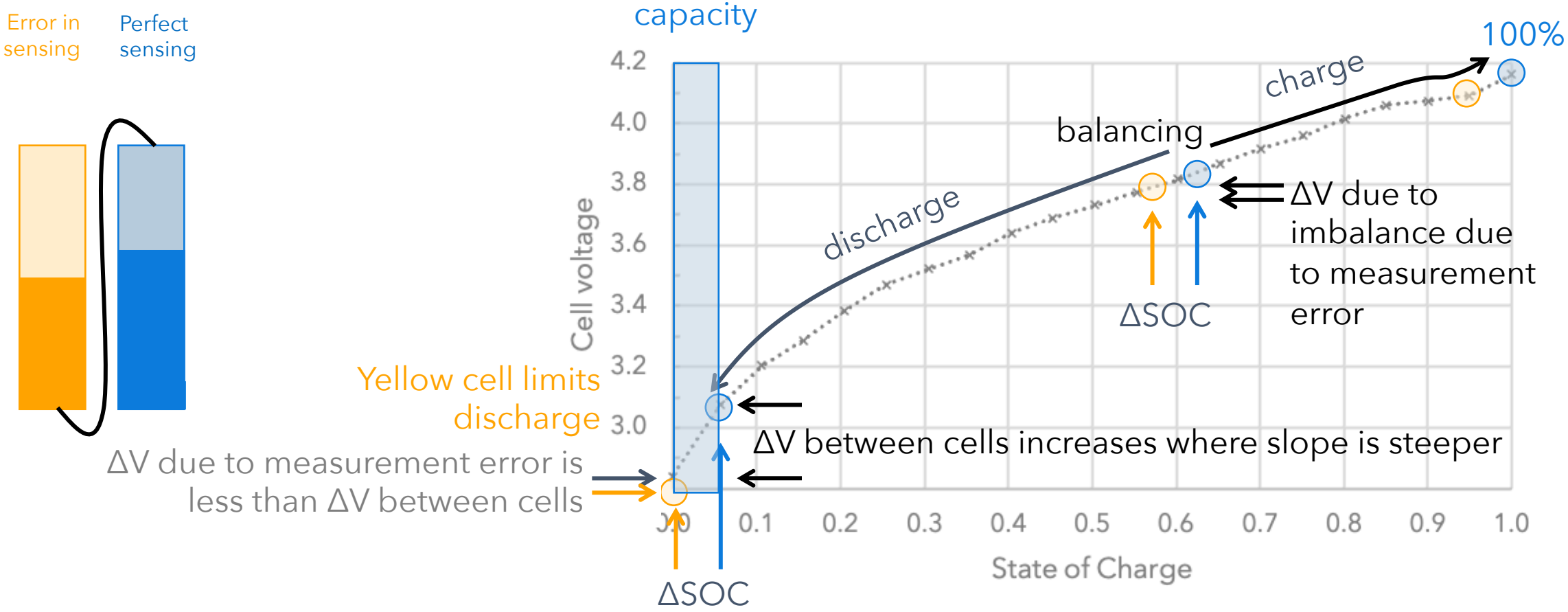
Summary of balancing & accuracy



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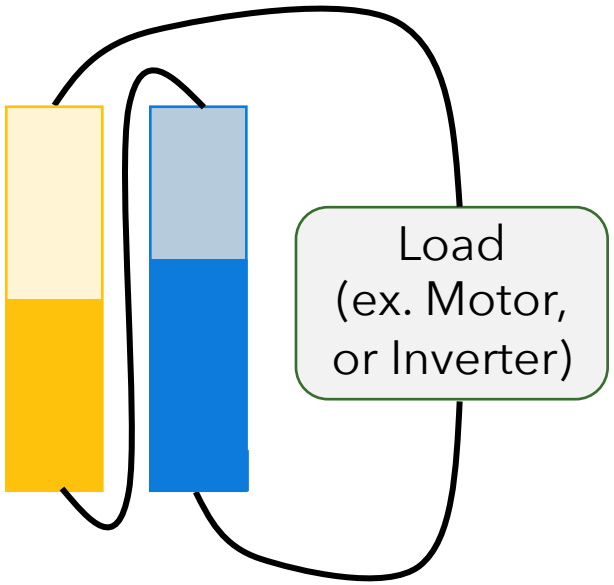
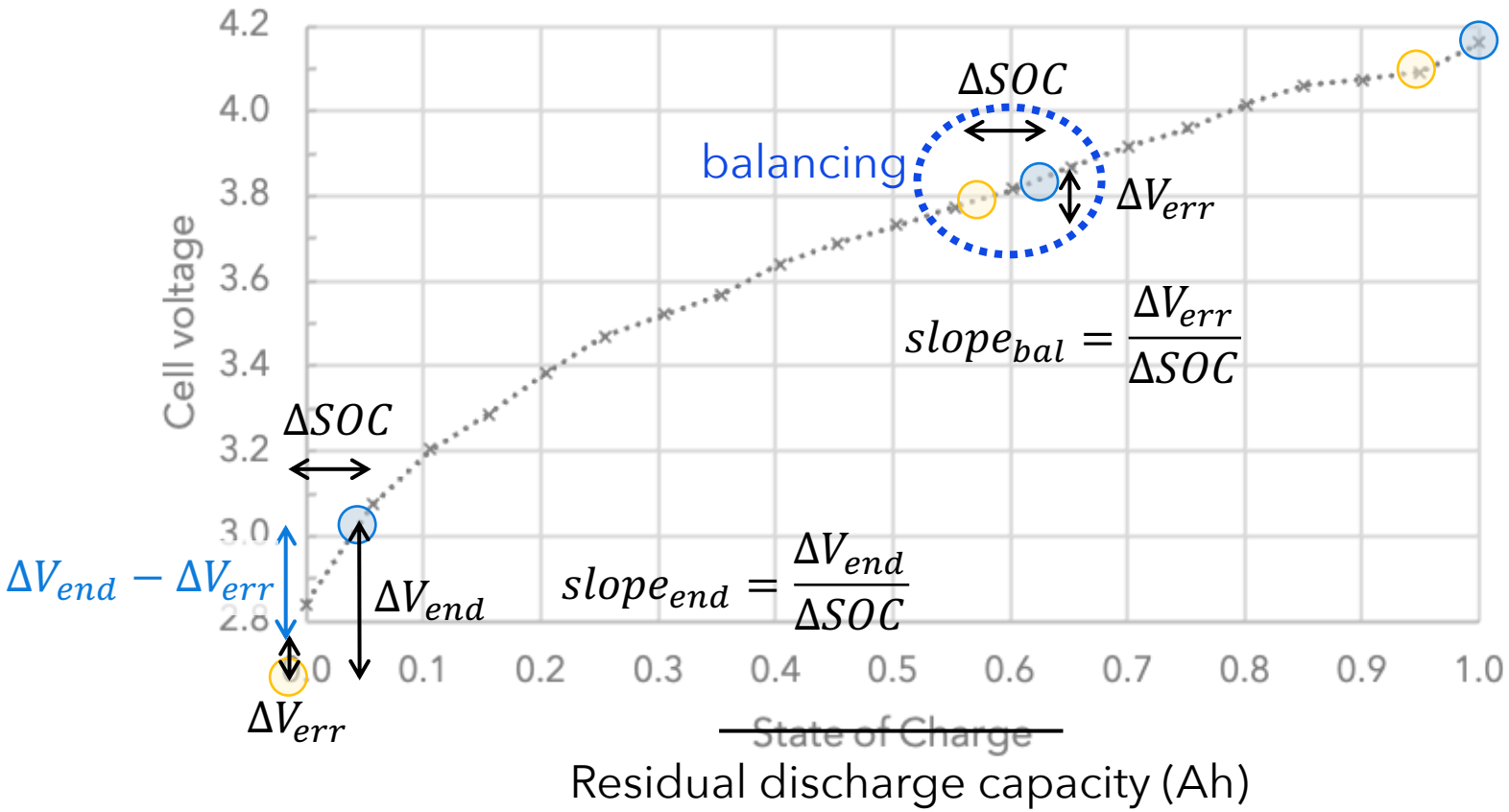
Concept, in a nutshell





If measurement error has a gain term:

$$\Delta V_{err}^{bottom} < \Delta V_{err}^{top}$$

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Equations

Concept → Equation → Implementation / Understanding

Goal:

ΔQ_{lost}=Capacity above 0% SOC at which blue cell ends discharge

Imbalance: $\Delta Q = \dfrac{\Delta V_{err}^{bal}}{slope_{bal}}$

$$\Delta V_{end} = slope_{end} \times \Delta Q$$

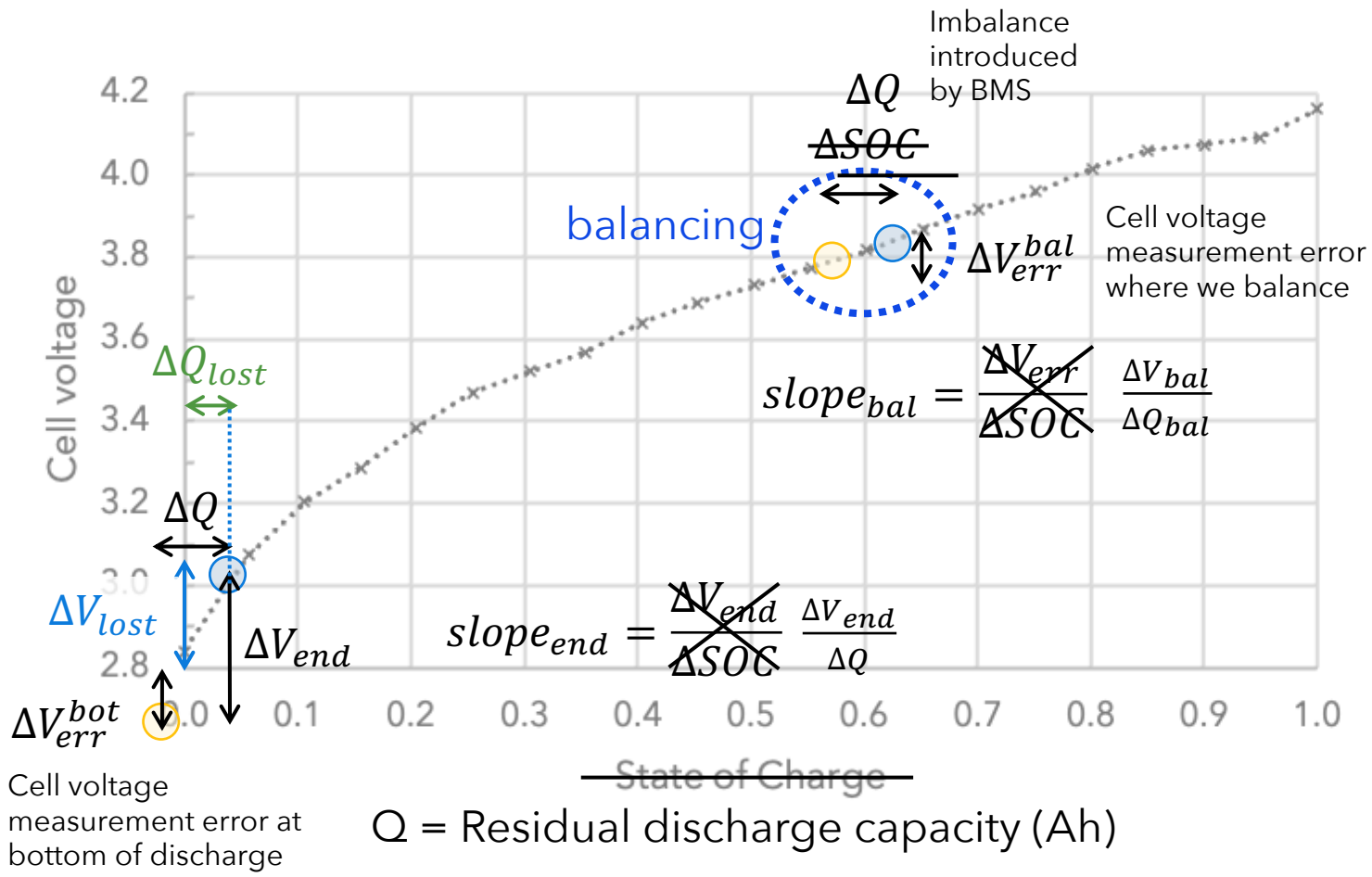
$$\Delta V_{end} = slope_{end} \times \left(\frac{\Delta V_{err}^{bal}}{slope_{bal}}\right)$$

$$\Delta V_{end} = \Delta V_{err}^{bal} \times \left(\frac{slope_{end}}{slope_{bal}}\right)$$

ΔV_{lost}= Voltage above 0% SOC at which blue cell ends discharge

$$\Delta V_{lost} = \Delta V_{end} - \Delta V_{err}^{bot}$$

$$\Delta Q_{lost} = \frac{\Delta V_{lost}}{slope_{end}}$$



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ΔV_{lost}= Voltage above 0% SOC at which blue cell ends discharge

$$\Delta V_{lost} = \Delta V_{end} - \Delta V_{err}^{bot}$$

$$\Delta Q_{lost} = \frac{\Delta V_{lost}}{slope_{end}}$$

$$\Delta Q_{lost} = \frac{\Delta V_{lost}}{slope_{end}} = \frac{\Delta V_{end} - \Delta V_{err}^{bot}}{slope_{end}}$$

$$\boxed{\Delta Q_{lost} = \frac{\Delta V_{err}^{bal} \times \left(\frac{slope_{end}}{slope_{bal}}\right) - \Delta V_{err}^{bot}}{slope_{end}}}$$

Capacity loss increases as the difference in slopes increases.

If OCV-SOC curve was linear (constant slope) and measurement error was constant, no capacity loss would occur due to imbalance:

If $slope_{bal} = slope_{end}$ and $\Delta V_{err}^{bal} = \Delta V_{err}^{bot}$

$$\Delta Q_{lost} = \frac{\Delta V_{err} \times \left(\frac{slope}{slope}\right) - \Delta V_{err}}{slope} = \frac{\Delta V_{err} - \Delta V_{err}}{slope} = 0$$

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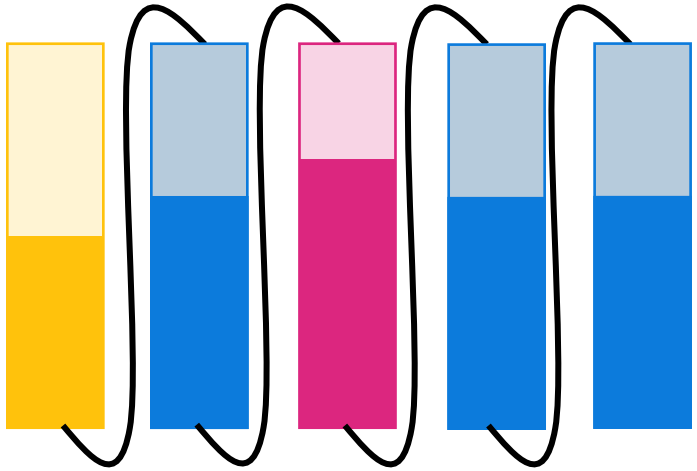
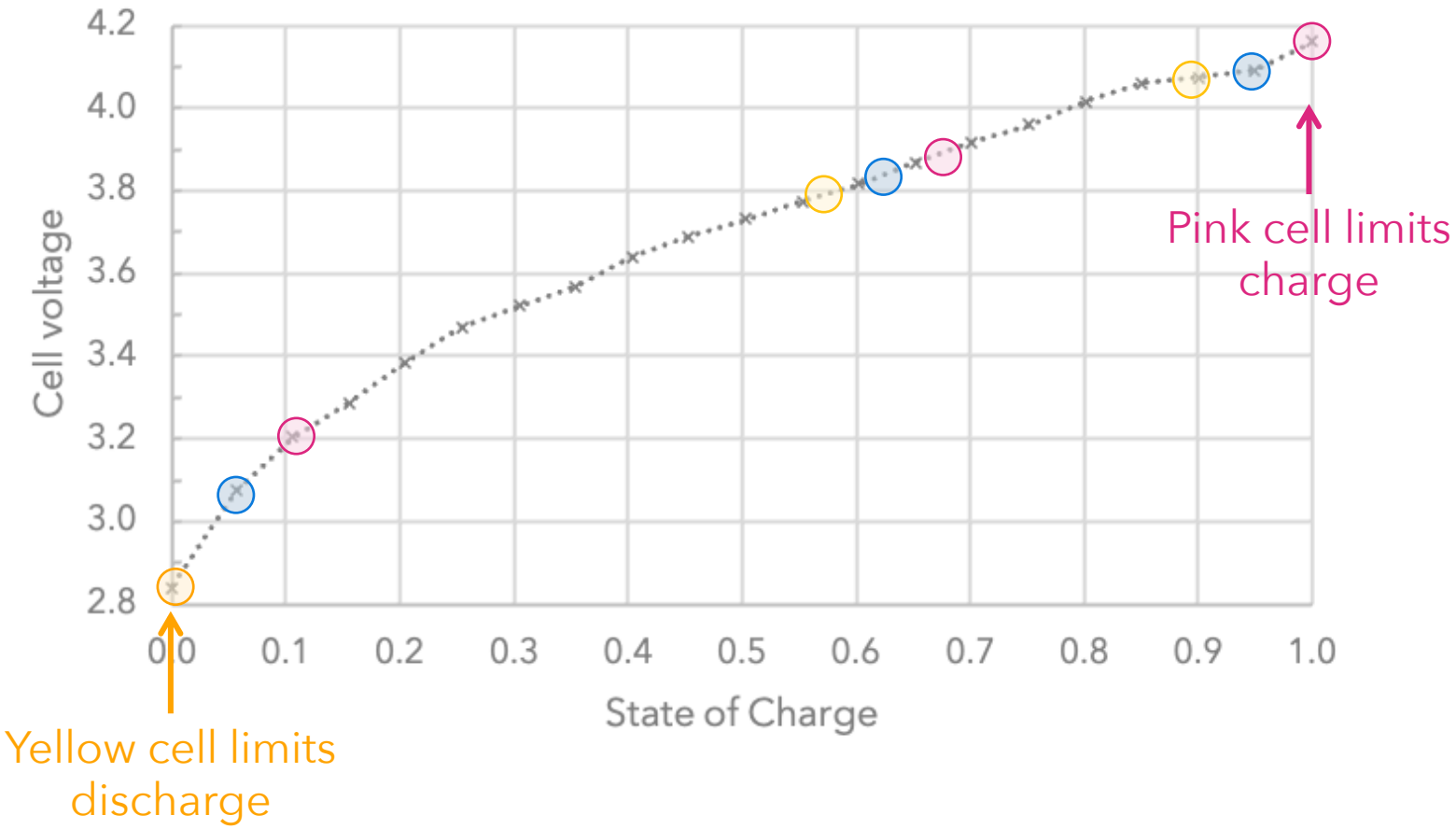
With more cells

Can generalize to:
- Both charge & discharge
- More cells

Roughly **double** possible capacity loss
(similar effect on top as we saw at bottom)





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Example of value of accuracy in $

The single-ended equation/analysis

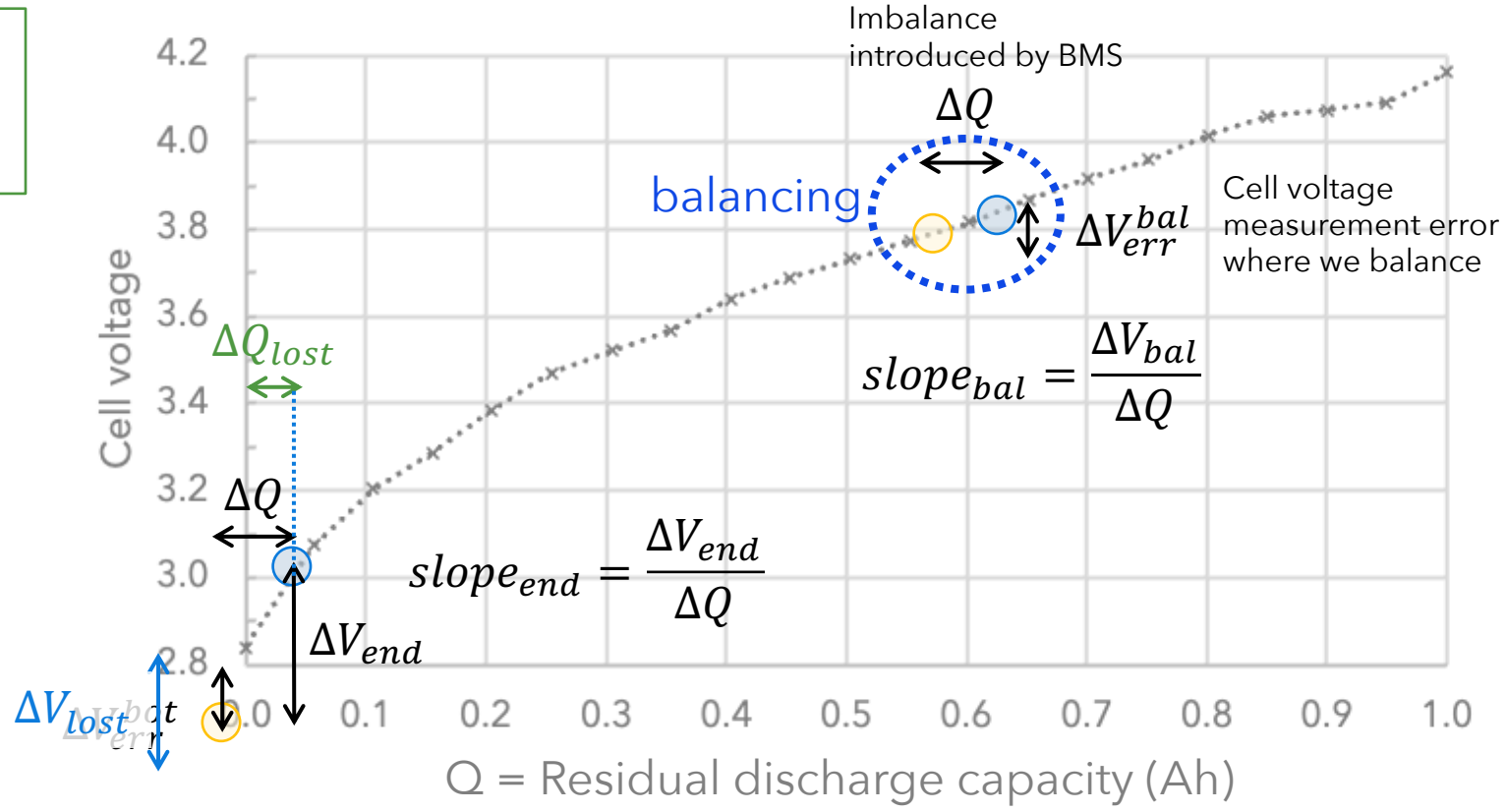
$$\Delta Q_{lost} = \frac{\Delta V_{err}^{bal} \times \left(\frac{slope_{end}}{slope_{bal}}\right) - \Delta V_{err}^{bot}}{slope_{end}}$$

What's the cost of this unused capacity?

- Q_{total} = total capacity target
- X_{total} = total cell cost in $
- $X_{lost} = \frac{Q_{lost}}{Q_{total}} \times X_{tot}$

Example:

- ~~50mV → 3.6% unused~~
- 10mV → 0.7% unused
- For $10,000 car battery → $70
- For $300 skateboard battery → $2.10



$7/mV is sensible to spend for more accuracy

21¢/mV is sensible to spend for more accuracy

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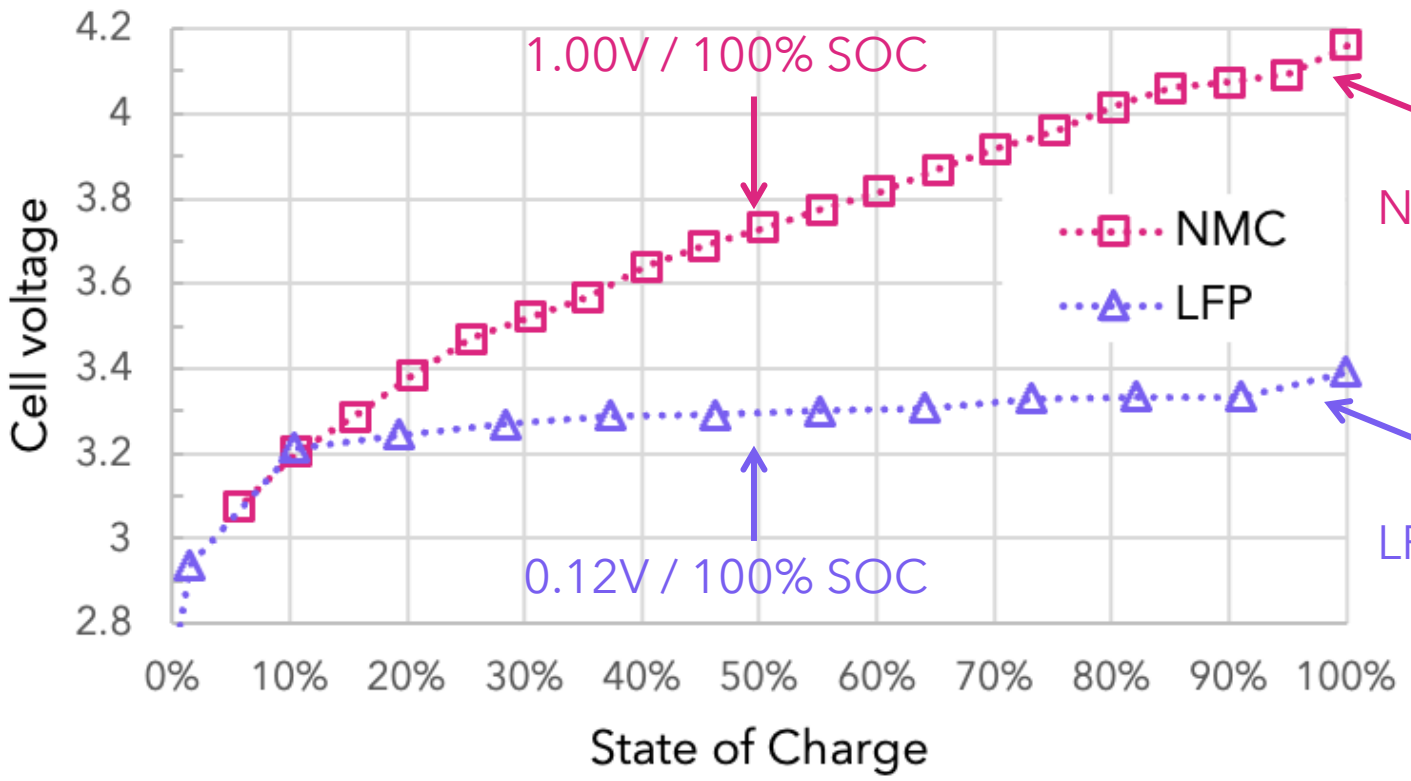
Value of accuracy varies across batteries

For which cell is accuracy more valuable?



1.42V / 100% SOC

NMC $/mV = $\frac{[battery\ cost\ in\ \$]}{[X]}$

0.67V / 100% SOC

LFP $/mV = $\frac{[battery\ cost\ in\ \$]}{\left[\frac{1}{2}X\right]} = \mathbf{2}\times[NMC\ \$/mV]$

$ per mV = $\frac{[battery\ cost\ in\ \$]}{\left[slope\ in\ \frac{mV}{\%SOC}\right]}$

Mid-range:
LFP $/mV = $\mathbf{8}\times[NMC\ \$/mV]$

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https://ofb.net/~ania/Anias-BMS-course/

Summary

We learned:

- Utility of common target, like $/mV or $/kg or $/A
- Identified where mV accuracy affects cost in $
- Derived equation for capacity lost (Q_{lost}) due to measurement error affecting balancing
- Calculated a $/mV accuracy target for two example systems (car & skateboard)

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Value of Accuracy : Video 2

Video 1
Video 2
Video 3
Video 4

- Syst s in $
- Cell
  - R
  - C
  - A
- Batt

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Value of Accuracy : Video 3

Video 1
- Systems engineering: Quantifying design variables in $

Video 2
- Cell voltage measurement accuracy ($/mV)
  - Review
  - Calculate value in $
    - 1st order
    - Better analysis

Video 3
  - Accuracy on BMS chip datasheets

Video 4
- Battery current measurement accuracy ($/A)

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Two vendors, two styles

bq79616

TEXAS INSTRUMENTS

BQ79616-Q1, BQ79614-Q1, BQ79612-Q1
16S/14S/12S Battery Monitor, Balanc

1 Features

- Qualified for automotive applications
- AEC-Q100 Qualified with the following results:
  - Device temperature grade 1: –40°C to +125°C ambient operating temperature range
  - Device HBM ESD classification level 2
  - Device CDM ESD classification level C4B
- Functional Safety-Compliant
  - Developed for functional safety applications
  - Documentation to aid ISO 26262 system design
  - Systematic capability up to ASIL D
  - Hardware capability up to ASIL D
- +/- 1.5mV ADC accuracy

LTC6813

ANALOG DEVICES

18-Ce

FEATURES

- Measures Up to 18 Battery Cells in Series
- 2.2mV Maximum Total Measurement Error
- Stackable Architecture for High Voltage Systems
- Built-In isoSPI™ Interface
  - 1Mb Isolated Serial Communications
  - Uses a Single Twisted Pair, Up to 100 Meters
  - Low EMI Susceptibility and Emissions
  - Bidirectional for Broken Wire Protection
- 290µs to Measure All Cells in a System
- Synchronized Voltage and Current Measurement
- 16-Bit Delta-Sigma ADC with Programmable Noise Filter

From: https://www.ti.com/product/BQ79616-Q1

From: https://www.analog.com/en/products/ltc6813-1.html

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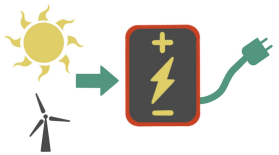
Electrical Characteristics Table



bq79616 ±1.5mV ADC accuracy

8.5 Electrical Characteristics (continued)

over operating -40°C to 125°C free-air temperature range, VBAT = 9V to 80V (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
ADC Accuracy						
I_{VC_DELTA}	VCn to VCn-1 input current delta (when Main ADC is on)	T_A = -20°C to 65°C			1.8	µA
		T_A = -40°C to 105°C			2	µA
I_{VC}	VCn input current (when Main ADC is on)			8	12	µA
R_{CB_INPUT}	CB pin input impedance (when AUX ADC is on)			16		MΩ
$V_{ACC_MAIN_CELL}$	Total channel accuracy for main ADC VCELL measurement, LPF_VCELL[2:0] = 0x03 setting;	2V<V_{CELL}<4.5V; T_A=25°C	-2.2		1.5	mV
		2V<V_{CELL}<4.5V; -20°C<T_A<65°C	-3.0		2.4	mV
		2V<V_{CELL}<4.5V; -40°C<T_A<105°C	-3.5		2.6	mV
		2V<V_{CELL}<4.5V; -40°C<T_A<125°C	-3.5		2.6	mV
		1V<V_{CELL}< 5V; -40°C<T_A<125°C	-3.7		2.8	mV
		-2V<V_{CELL}< 5V; -40°C<T_A<125°C	-4.5		3.2	mV

LTC6813 2.2mV Maximum Total Measurement Error

ELECTRICAL CHARACTERISTICS The ● denotes the specifications which apply over the full specified temperature range, otherwise specifications are at T_A = 25°C. The test conditions are V^+ = 59.4V, V_{REG} = 5.0V unless otherwise noted. The ISOMD pin is tied to the V^- pin, unless otherwise noted.

SYMBOL	PARAMETER	CONDITIONS		MIN	TYP	MAX	UNITS
ADC DC Specifications							
	Measurement Resolution				0.1		mV/Bit
	ADC Offset Voltage	(Note 2)			0.1		mV
	ADC Gain Error	(Note 2)			0.01		%
	Total Measurement Error (TME) in Normal Mode	C(n) to C(n–1), GPIO(n) to V^- = 0			±0.2		mV
		C(n) to C(n–1) = 2.0				±1.6	mV
		C(n) to C(n–1), GPIO(n) to V^- = 2.0, LTC6813I	●			±1.8	mV
		C(n) to C(n–1), GPIO(n) to V^- = 2.0, LTC6813H	●			±2.0	mV
		C(n) to C(n–1) = 3.3				±2.2	mV
		C(n) to C(n–1), GPIO(n) to V^- = 3.3, LTC6813I	●			±3.0	mV
		C(n) to C(n–1), GPIO(n) to V^- = 3.3, LTC6813H	●			±3.3	mV
		C(n) to C(n–1) = 4.2				±2.8	mV
		C(n) to C(n–1), GPIO(n) to V^- = 4.2, LTC6813I	●			±3.8	mV
		C(n) to C(n–1), GPIO(n) to V^- = 4.2, LTC6813H	●			±4.2	mV
		C(n) to C(n–1), GPIO(n) to V^- = 5.0			±1		mV

Battery: 0°C to 45°C
PCB: 0°C to ~100°C

Battery: 2V to 4.2V

±1.5mV and 2.2mV are "typical" values.

3.5mV and 3.8mV cover operating temperature range.

From: https://www.ti.com/product/BQ79616-Q1

From: https://www.analog.com/en/products/ltc6813-1.html

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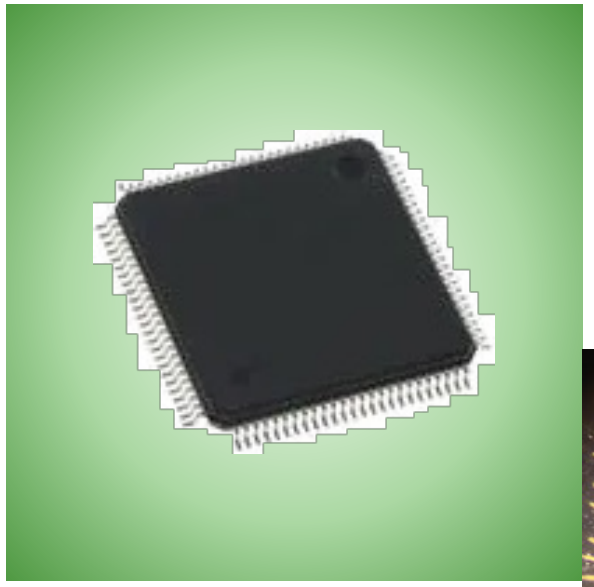
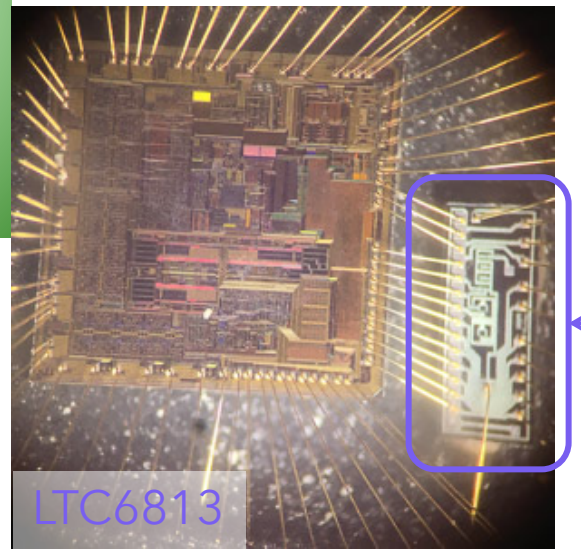
Source of error: Soldering

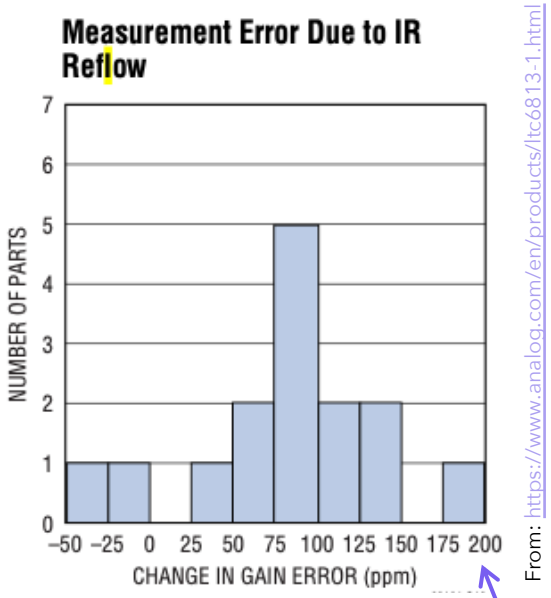
Mechanically-induced error:

- Soldering
- Long-term drift





LTC6813



200ppm: $\frac{200}{1{,}000{,}000} \times 4.2V = 0.84mV$

Buried Zener reference die

More about LTC's approach to references: https://www.analog.com/media/en/technical-documentation/application-notes/an82f.pdf

bq79616

???

2mV ?

5mV ?

15mV ?

30mV ?

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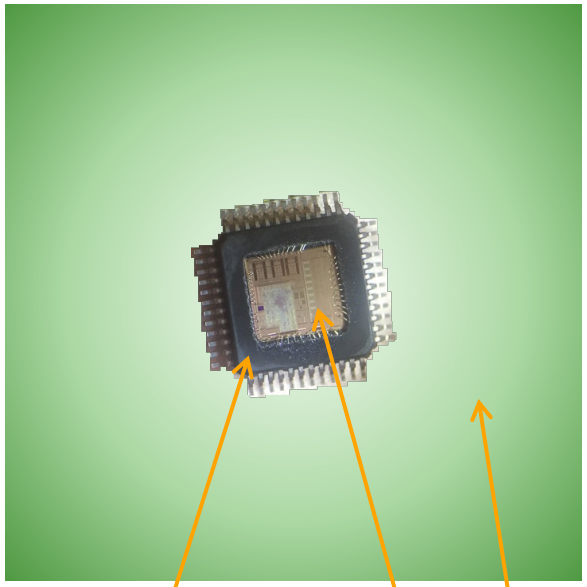
Source of error: Long term drift



Epoxy and silica chip package

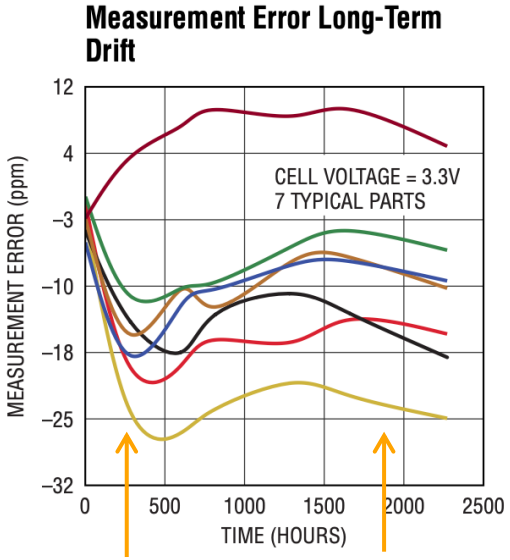
Silicon die

Fiberglass PCB (printed circuit board)

Different coefficients of thermal expansion

LTC6811 BMS chip

From: https://www.analog.com/en/products/ltc6811-1.html



Rapid change in first week or two

Slower subsequent drift

Drift is worse in chips without a buried Zener reference or other mitigation

REF34 and REF 50 references

From: https://www.ti.com/lit/pdf/sbaa436

Nice document

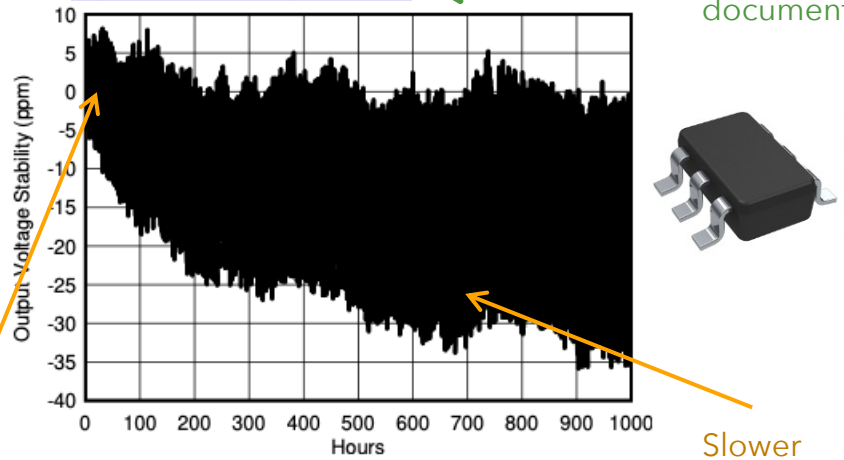


Figure 4-3. REF34 - SOT23-3 Package LTD

More change in first week

Slower subsequent drift

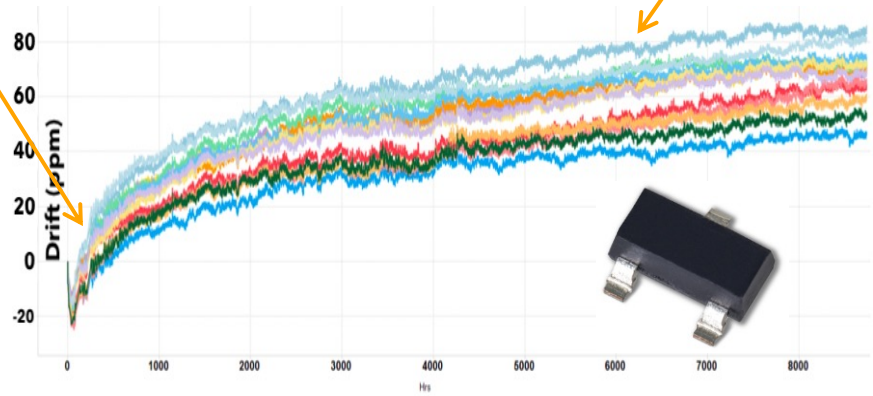


Figure 6-1. REF50 - VSSOP Package LTD 35°C

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Speculate…

Table 3-1. REF34-Q1 SOT23-3 Error Breakdown

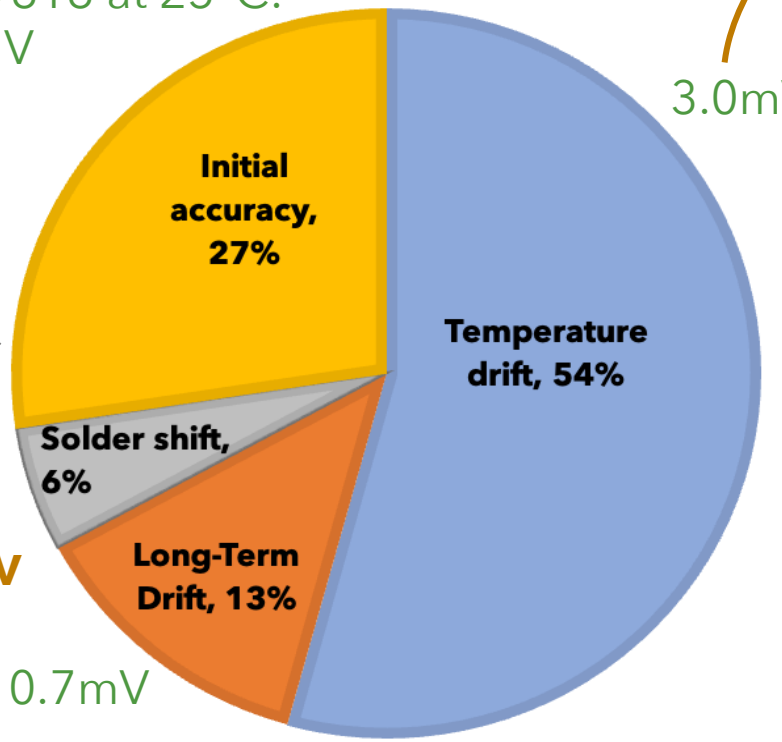
Parameter	Error (ppm)	Source	Notes
Initial accuracy	500	Data sheet maximum value	
Solder shift	100	Data sheet typical value	
Temperature drift	990	Data sheet maximum value	6ppm/C × (125 – (-40))°C = 990 ppm
Long-Term Drift	234	Data sheet typical value	Conservative estimate based on 10 yr value

From: https://www.ti.com/lit/pdf/sbaa436

Estimate: 1.5mV + 3.0mV = **4.5mV**
bq79616 datasheet: **3.5mV** for -40°C to 125°C

bq79616 at 25°C: 1.5mV

3.0mV



0.3mV

0.3mV + 0.7mV = **1.0mV**

0.7mV

bq79616 is a much larger chip, so effects due to mechanical stresses may also be larger.

10mm

3mm

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Two vendors, two styles

bq79616

TEXAS INSTRUMENTS

BQ79616-Q1, BQ79614-Q1, BQ79612-Q1
16S/14S/12S Battery Monitor, Balanc

1 Features

- Qualified for automotive applications
- AEC-Q100 Qualified with the following results:
  - Device temperature grade 1: –40°C to +125°C ambient operating temperature range
  - Device HBM ESD classification level 2
  - Device CDM ESD classification level C4B
- Functional Safety-Compliant
  - Developed for functional safety applications
  - Documentation to aid ISO 26262 system design
  - Systematic capability up to ASIL D
  - Hardware capability up to ASIL D
- +/- 1.5mV ADC accuracy

3.5mV for -40°C to 125°C

+ ? >>1mV ? for solder shift and long-term drift

From: https://www.ti.com/product/BQ79616-Q1

LTC6813

ANALOG DEVICES

18-Ce

FEATURES

- **Measures Up to 18 Battery Cells in Series**
- **2.2mV Maximum Total Measurement Error**
- **Stackable Architecture for High Voltage Systems**
- **Built-In isoSPI™ Interface**
  - 1Mb Isolated Serial Communications
  - Uses a Single Twisted Pair, Up to 100 Meters
  - Low EMI Susceptibility and Emissions
  - Bidirectional for Broken Wire Protection
- 290µs to Measure All Cells in a System
- Synchronized Voltage and Current Measurement
- 16-Bit Delta-Sigma ADC with Programmable Noise Filter

3.8mV for -40°C to 125°C

And that's all.

From: https://www.analog.com/en/products/ltc6813-1.html

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Third vendor

TLE9012

Aims to address both long-term drift and solder shift

- High-accuracy measurement for SoC and SoH calculation
- Integrated stress sensor with digital compensation algorithm and temperature-compensated measurements

8.2 Electrical characteristics primary cell voltage measurement (PCVM)

Table 9 Electrical characteristics

$V_{VS} = V_{VS_functional}$, T_j = -40°C to +150°C, all voltages with respect to GND, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note or condition	P-Number
		Min.	Typ.	Max.			
PCVM accuracy EoL - 5	$PCVM_{ERR_EOL_5}$	-4.1	–	4.1	mV	13) 14)11) 1. 16-bit mode 2. $1\ V \le (V_{Un+1} - V_{Un}) \le 3.6\ V$ 3. $-40°C \le T_j \le 150°C$ 4. ±3 sigma distribution within absolute minimum and maximum limits	PRQ-607
PCVM accuracy EoL - 6	$PCVM_{ERR_EOL_6}$	-4.5	–	4.5	mV	13) 14)11) 1. 16-bit mode 2. $3.6\ V < (V_{Un+1} - V_{Un}) \le 4.3\ V$ 3. $-40°C \le T_j \le 150°C$ 4. ±3 sigma distribution within absolute minimum and maximum limits	PRQ-608

13 Lower resolution has additional quantization error e.g. additional $PCVM_{ERR_EOL}$ ± 2 LSB[m]; m bits: 14 ≤ m ≤ 15

Please contact Infineon for more details for other ADC resolutions.

14 End-of-Life (EoL) accuracy; according to AEC-Q100 Grade 1 Rev. H automotive qualification

Assesses long-term drift

11 With 12 cells attached and activated

Recall:

bq79616

3.5mV for -40°C to 125°C

+ ? >1mV ? for solder shift and long-term drift

→ **>4.5mV** total

LTC6813

3.8mV for -40°C to 125°C

That's all.

TLE9012

4.5mV for -40°C to 150°C

That's all.

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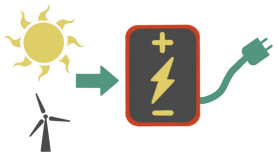
Compare all three



	bq79616	LTC6813	TLE9012
First page	**±1.5mV** ADC accuracy	**2.2mV** Maximum Total Measurement Error	High-accuracy
Full range with temperature, solder shift, long-term drift	**3.5mV** for -40°C to 125°C **+ ? >>1mV ?** for solder shift and long-term drift **>4.5mV ???** total	**3.8mV** for -40°C to 125°C That's all.	**4.5mV** for -40°C to 150°C That's all.
Mitigations For mechanical stresses	? none ?	Buried Zener reference.	Integrated stress sensor

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What is sigma?

TLE9012

8.2 Electrical characteristics primary cell voltage measurement (PCVM)

Table 9 Electrical characteristics

$V_{VS} = V_{VS_functional}$, T_j = -40°C to +150°C, all voltages with respect to GND, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note or condition	P-Number
		Min.	Typ.	Max.			
PCVM accuracy EoL - 5	$PCVM_{ERR_EOL_5}$	-4.1	–	4.1	mV	*13) 14)11)* **1.** 16-bit mode **2.** 1 V ≤ (V_{Un+1} - V_{Un}) ≤ 3.6 V **3.** -40°C ≤ T_j ≤ 150°C **4.** ±3 sigma distribution within absolute minimum and maximum limits	PRQ-607
PCVM accuracy EoL - 6	$PCVM_{ERR_EOL_6}$	-4.5	–	4.5	mV	*13) 14)11)* **1.** 16-bit mode **2.** 3.6 V < (V_{Un+1} - V_{Un}) ≤ 4.3 V **3.** -40°C ≤ T_j ≤ 150°C **4.** ±3 sigma distribution within absolute minimum and maximum limits	

3 of 1000 units may have greater error at some corners of operating range

13 Lower resolution has additional quantization error e.g. additional $PCVM_{ERR_EOL}$ ± 2 m ≤ 15

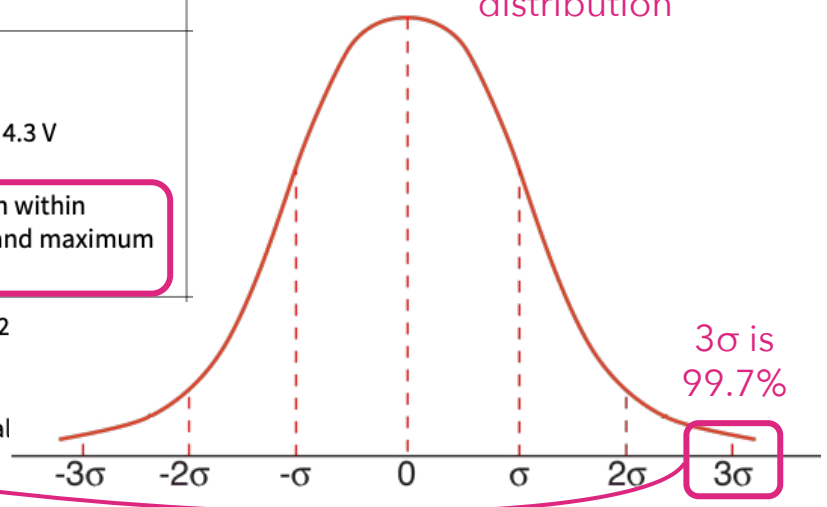
Please contact Infineon for more details for other ADC resolutions.

14 End-of-Life (EoL) accuracy; according to AEC-Q100 Grade 1 Rev. H automotive qual

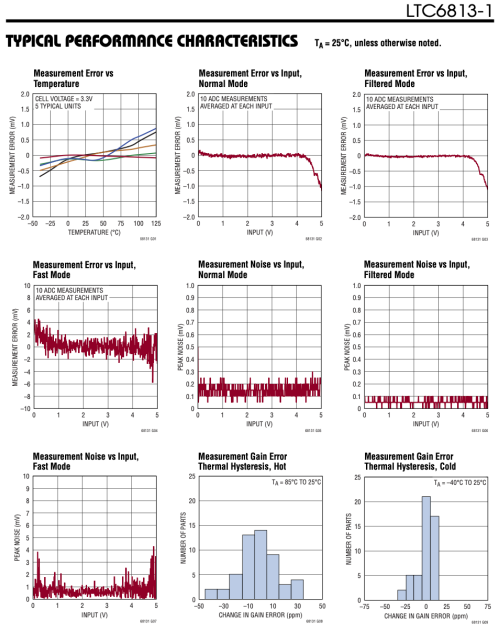
11 With 12 cells attached and activated

- Input voltage
- Temperature variation
- Power supply rejection
- Noise rejection
  - Chip's internal noise probably included
  - Rejection of external noise probably not included
- Mismatch or part-to-part variation
- …

Gaussian distribution



3σ is 99.7%



From: https://www.analog.com/en/products/ltc6813-1.html

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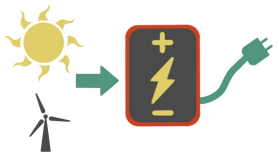
https://ofb.net/~ania/Anias-BMS-course/

Compare all three



	bq79616	LTC6813	TLE9012
First page	**±1.5mV** ADC accuracy	**2.2mV** Maximum Total Measurement Error	High-accuracy
Full range with temperature, solder shift, long-term drift	3.5mV for -40°C to 125°C + ? >>1mV ? for solder shift and long-term drift **>4.5mV ???** total	**3.8mV** for -40°C to 125°C That's all.	**4.5mV** for -40°C to 150°C That's all.
Mitigations For mechanical stresses	? none ?	Buried Zener reference.	Integrated stress sensor
Applies to what fraction of units?	**?? %** of parts	**?? %** of parts	**3σ : ≥99.7%** of parts
Who's best?		Datasheets don't tell the whole story.	

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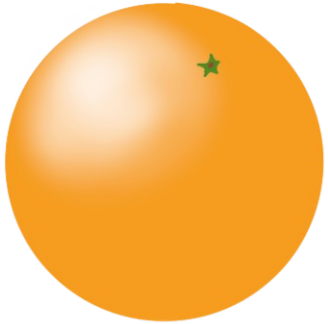
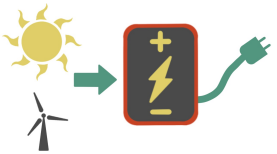
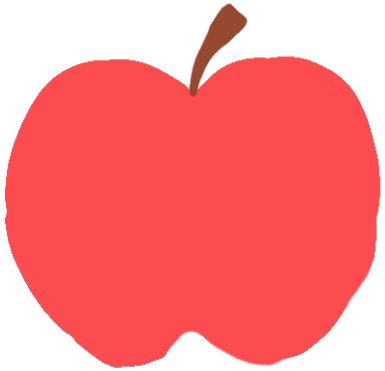
Summary of accuracy on datasheets







We learned:

- Front page accuracy value may not reflect actual performance
- Electrical Characteristic (EC) Table provides better information
- Mechanical stresses (solder shift and long-term drift) are significant for high-accuracy chips like BMS chips
- Whether and how mechanical stress is addressed by design and captured in datasheet values varies among vendors
- Manufacturing variations tend to be Gaussian, and vendors differ in transparency about how many standard deviations are assumed in datasheet values

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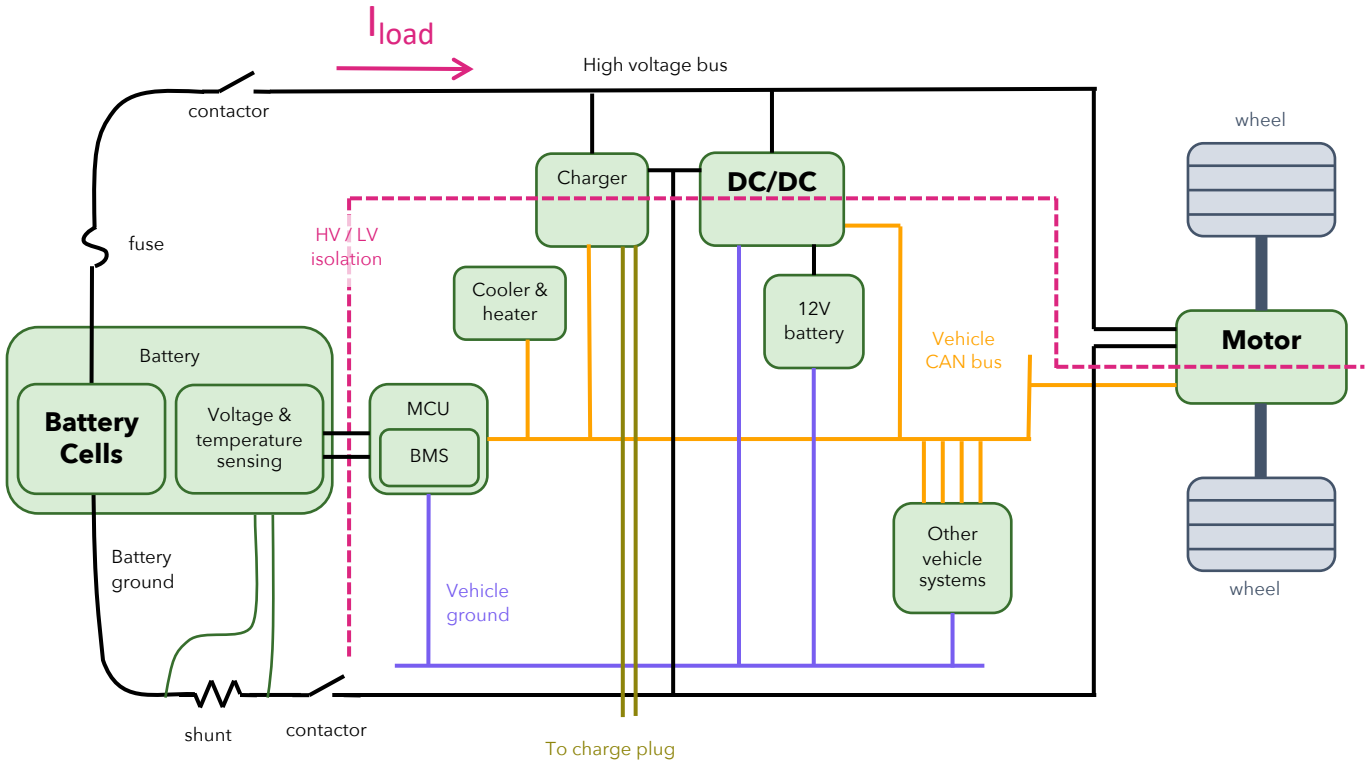
System noise

Resistive effects: $V_{battery} = V_{internal} - I_{load} \times R_{parasitic}$

Inductive effects: $V_{battery} = L_{parasitic} \times \frac{dI_{load}}{dt}$

Voltage ripple is system noise



bq79616

6.5Hz, 13Hz , 26Hz, 53Hz, 111Hz, 240Hz, 600Hz

LTC6813

422Hz, 1kHz, 2kHz, 3kHz, 14kHz

TLE9012

None?

RC filters outside chip, on PCB, have higher corner frequencies, thus providing less attenuation

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Value of Accuracy : Video 3

Video 1
- Syst s in $

Video 2
- Cell
  - R
  - C
  - A

Video 3

Video 4
- Batt

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Value of Accuracy : Video 4

Video 1
- Systems engineering: Quantifying design variables in $

Video 2
- Cell voltage measurement accuracy ($/mV)
  - Review
  - Calculate value in $
    - 1st order
    - Better analysis
  - Accuracy on BMS chip datasheets

Video 3

Video 4
- Battery current measurement accuracy ($/A)

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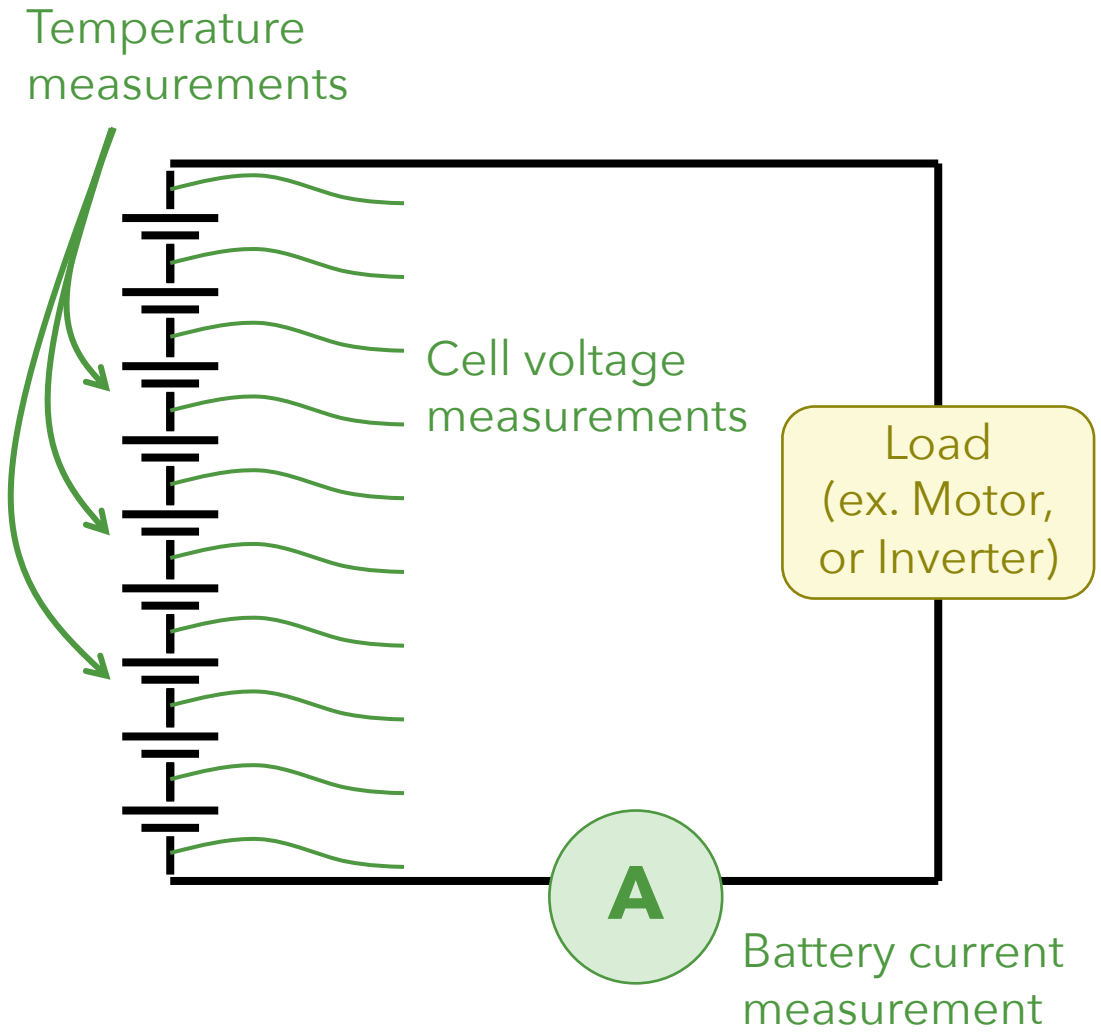
How we use battery current measurement



Cell voltage meas.

Temperature meas.

Battery current meas.

Let's look deeper

State estimation (SOC, SOH, power)

Fault detection (safety)

Firm specs

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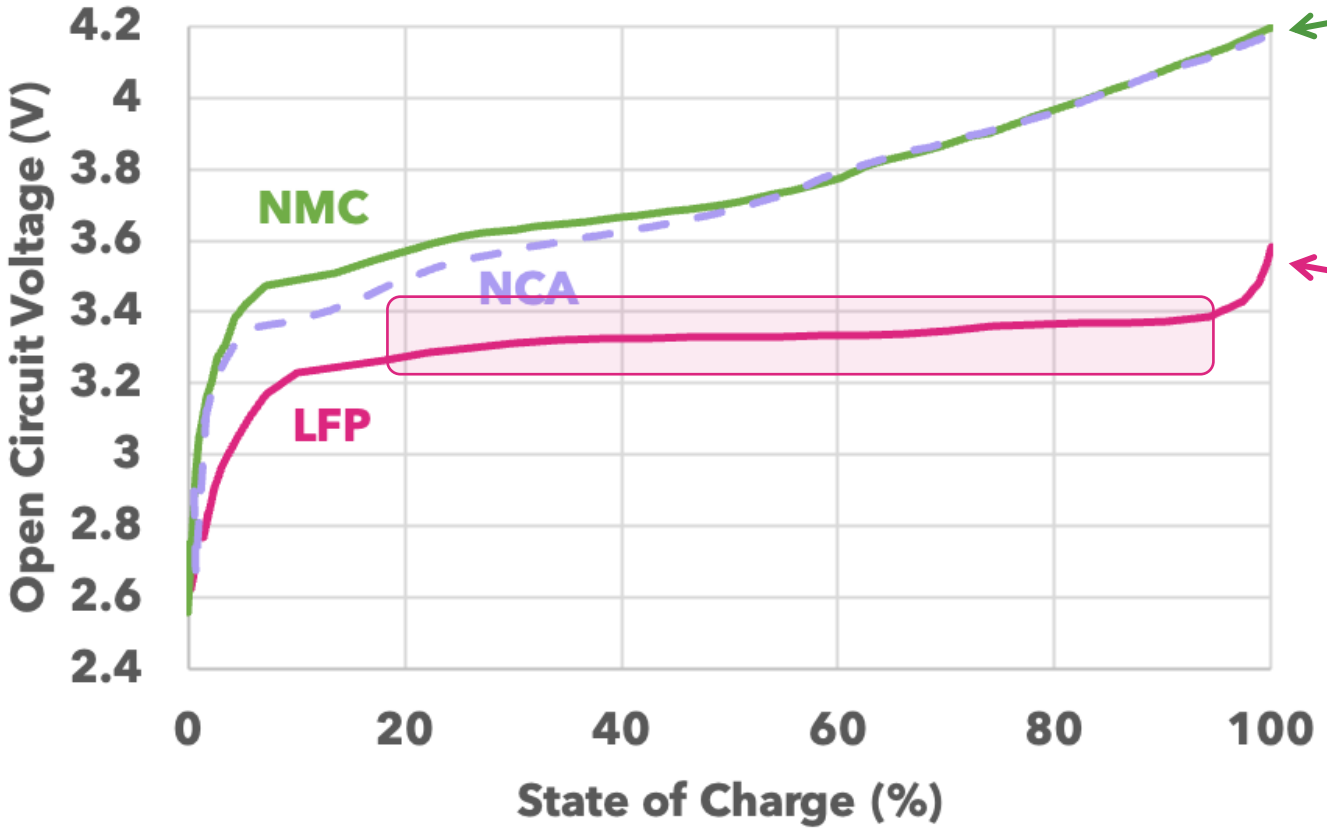
Battery current and SOC estimation



Data from Fig. 1 in DOI: 10.1149/2.0411609jes [Keil 2016]

- **Cell voltage** is good source of info when battery is rested, and limits charge/discharge.
- **Battery current** improves estimation accuracy when current is flowing.
- **Cell voltage** is provides little information for most of operating range.
- **Battery current** is critical for SOC estimation for most of operating range.

For both:

- **Cell voltage** determines charge/discharge limits
  - ➢ affects actual available capacity
- **Battery current** affects SOC estimation

Value of battery current measurement accuracy is greater for LFP than for NMC/NCA

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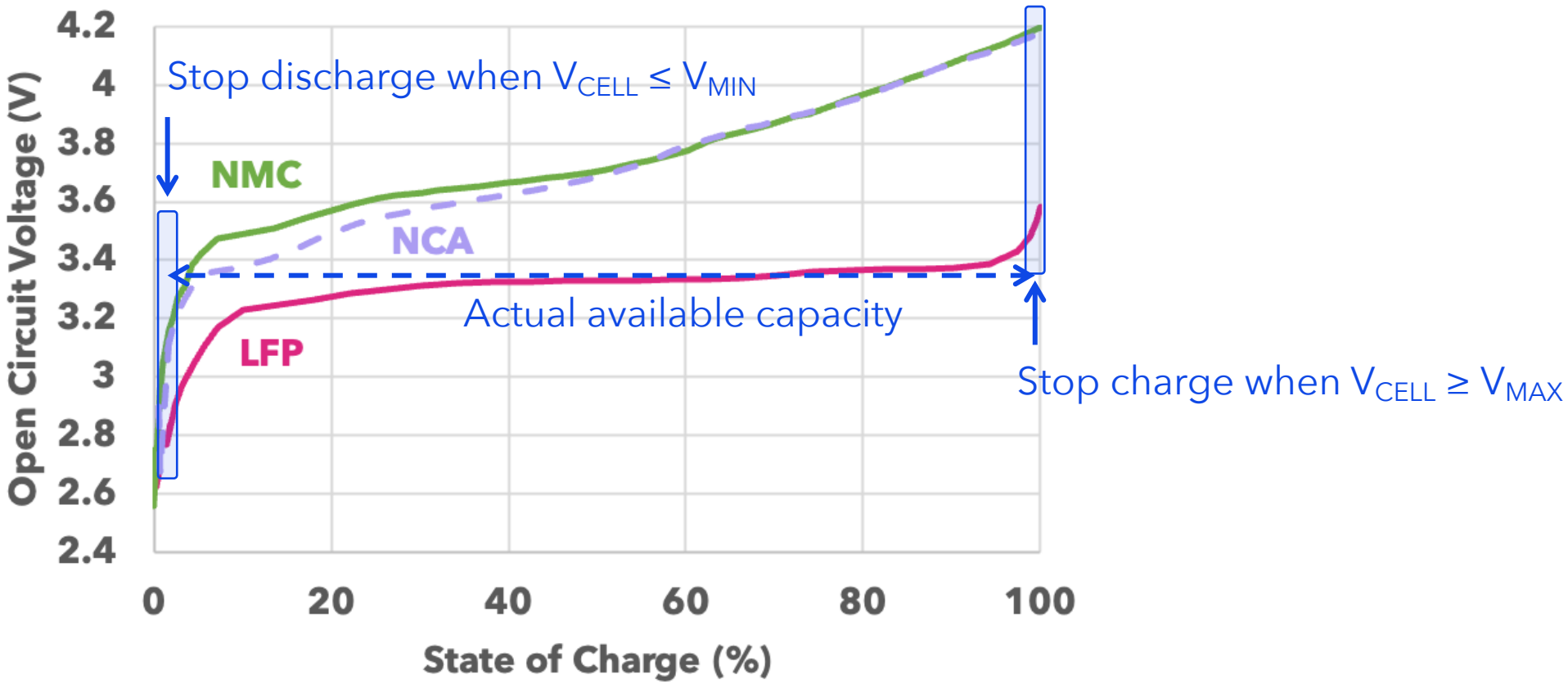
Cell voltage meas. → actual avail. capacity



Data from Fig. 1 in DOI: 10.1149/2.0411609jes [Keil 2016]

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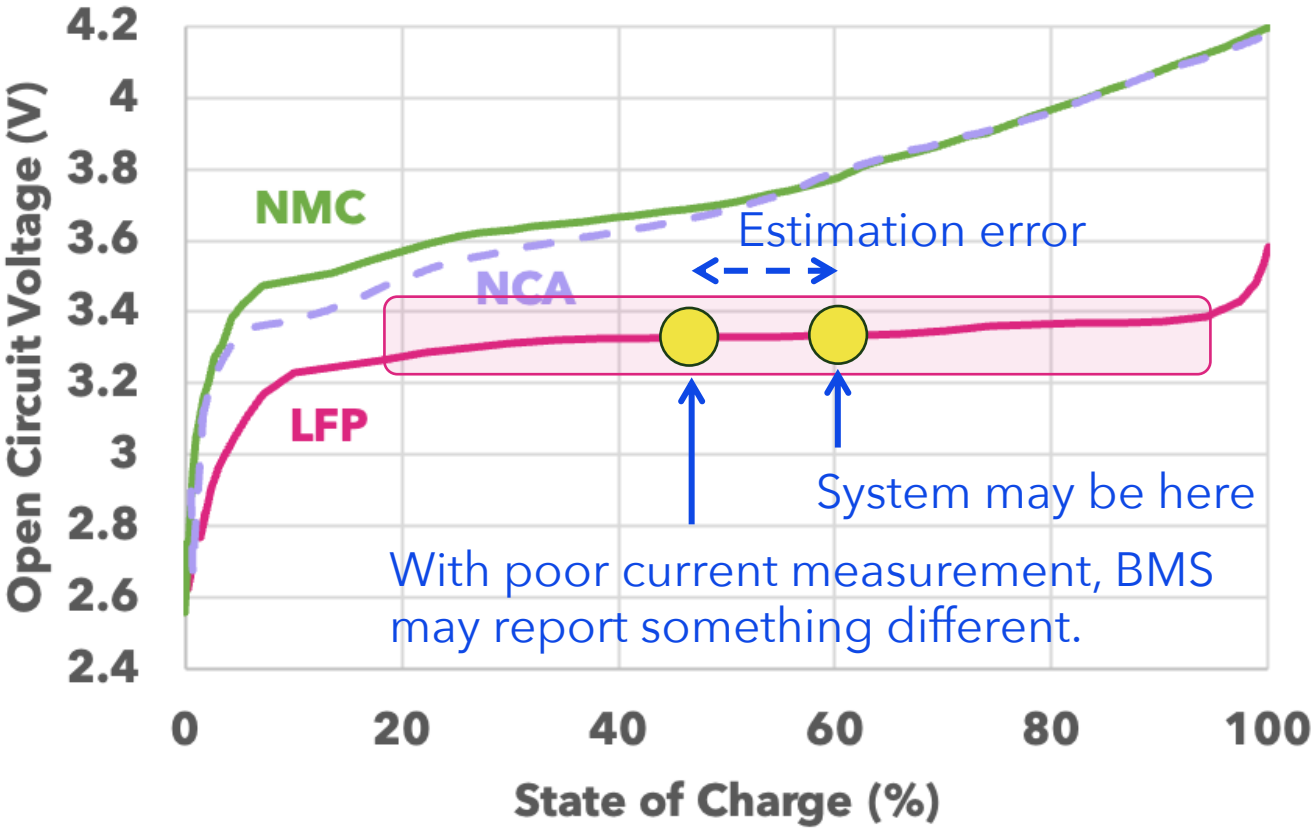
Battery current meas → SOC estimate error

Battery current measurement error affects the error between estimated and actual residual capacities.



Data from Fig. 1 in DOI: 10.1149/2.0411609jes [Keil 2016]

Does estimation error matter?

Application examples:

- **Car:** Important. Over-estimate may leave a driver stranded. Error looks bad to customers.
- **Home solar storage:** Moderate impact in supplemental system where solar panels complement grid. Error looks bad, and users often expect excellence of expensive systems.
- **E-assist bicycle:** Limited impact, since user has alternate way to propel vehicle (back-up energy source)

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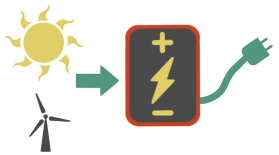
How to value current accuracy?



Current meas. → SOC estimate → $/mA

1) Model the system

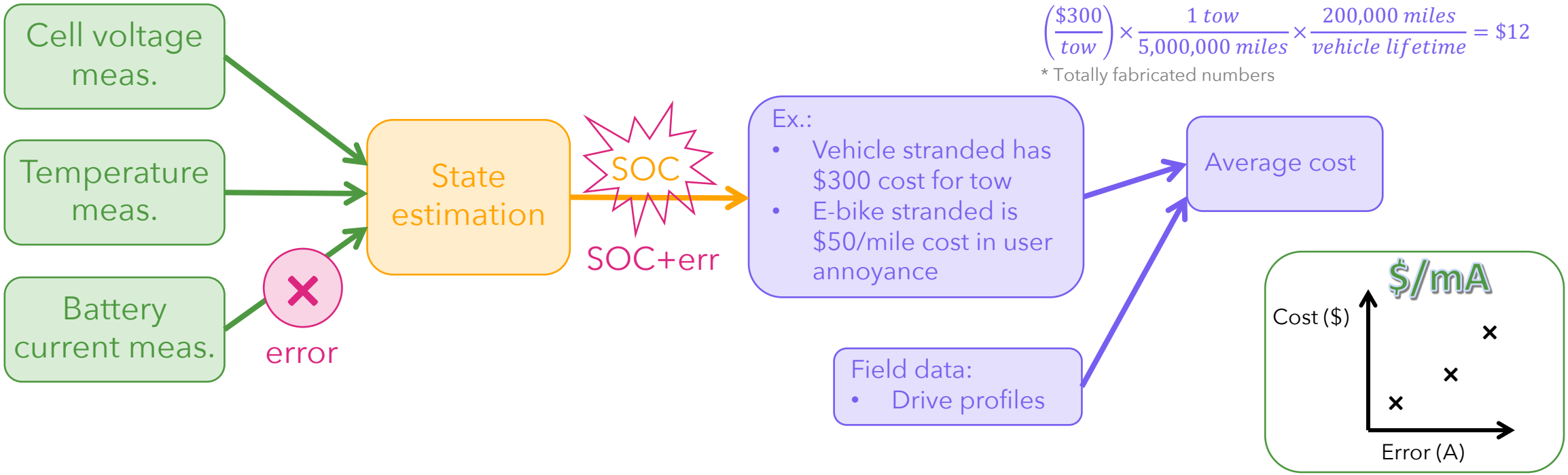
2) Sensitivity analysis

3) Tie SOC to a cost

4) Estimate frequency of occurrence and calculate average cost

$$\left(\frac{\$300}{tow}\right) \times \frac{1\ tow}{5{,}000{,}000\ miles} \times \frac{200{,}000\ miles}{vehicle\ lifetime} = \$12$$

* Totally fabricated numbers



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Sensitivity analysis…

…is tough early in design when hardware is under design.

1) Model the system

Software and algorithms are rarely ready before hardware (such as current sensor)

3) Tie SOC to a cost

Field data becomes available after product is in production

Cell voltage meas.

Temperature meas.

Battery current meas.

error

State estimation

SOC

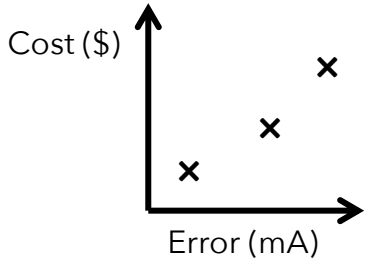
SOC+err

Ex.:
- Vehicle stranded has $300 cost for tow
- E-bike stranded is $50/mile cost in user annoyance

$$\left(\frac{\$300}{tow}\right) \times \frac{tow}{5{,}000\ miles} \times \frac{200{,}000\ miles}{vehicle\ lifetime} = \$12$$

Average cost

Field data:
- Drive profiles



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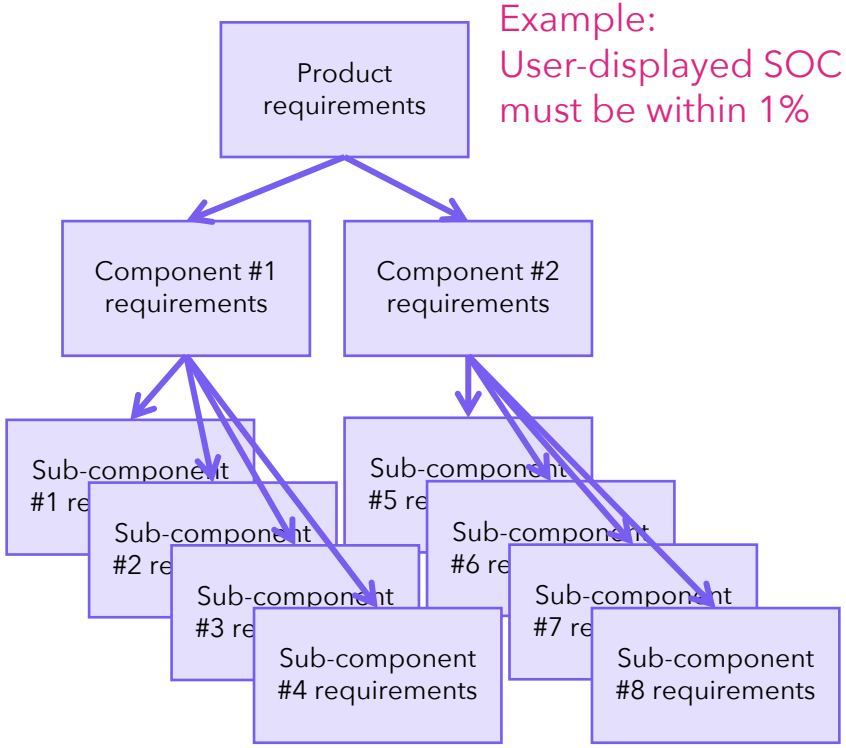
How to value current accuracy?

Traditional approach:

- Executive or product manager or system architect sets a target, based on intuition about the value of image and customer satisfaction
- Engineer designs accuracy goals correspondingly



Example:
User-displayed SOC estimate must be within 1%

Summary of value of current sense accuracy

Key take-aways:

- Current sensing matters most for SOC estimation, less for actual capacity
- Assigning value to an estimate is tricky
- Intuition-based top-down target may be appropriate for specifying current sense accuracy

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